

Student Name: _____

Class: _____ *Batch:* _____

Subject: _____

Lab course on Advance Communication (Semester - IV) (S.Y.B.Sc. Computer Science)



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S.Y.B.Sc. Computer Science***INDEX*****Lab course on Advance Communication****Course No. - CS-292-MNP**

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4.	Study of Binary Phase Shift Keying (BPSK)	
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6.	Study of Frequency Shift Keying (FSK)	
7.	Study of Time Division Multiplexing (TDM)	
8.	Error Detection and Correction using Hamming code	
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13.	Temperature and humidity sensing and monitoring using Arduino	

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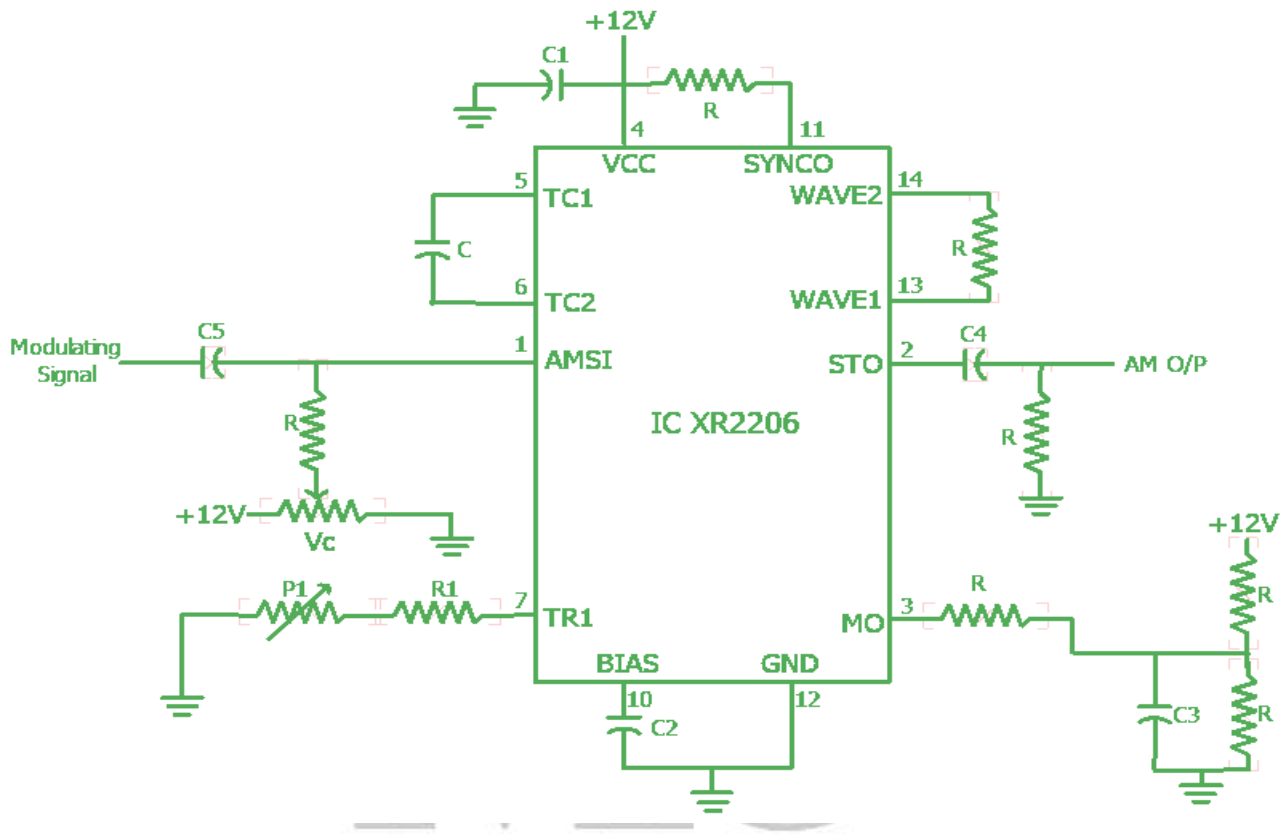
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Experiment – 1

Title: Study of Amplitude Modulation

Aim: To study Amplitude Modulation and observed the output.

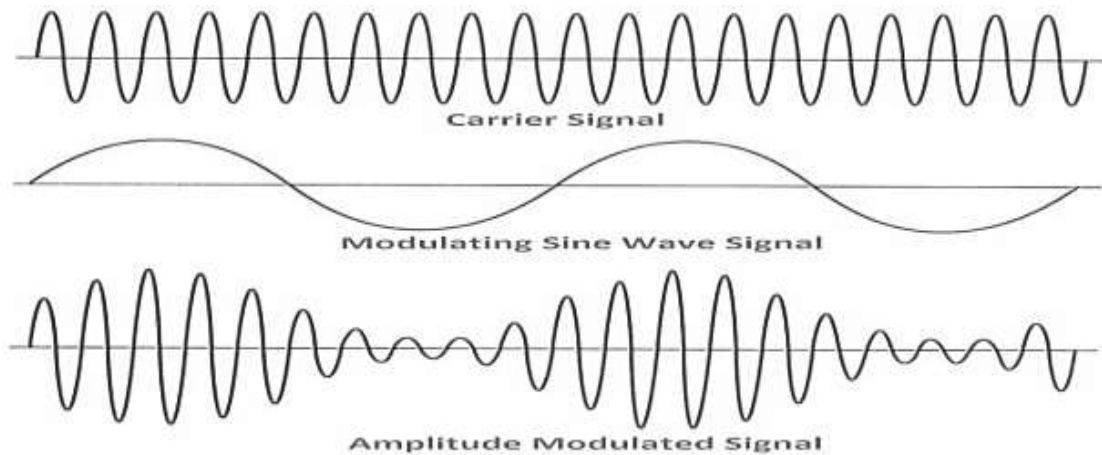
Circuit Diagram:



Theory:

Amplitude modulation is a process by which the wave signal is transmitted by modulating the amplitude of the signal. It is often called AM and is commonly used in transmitting a piece of information through a radio carrier wave. Amplitude modulation is mostly used in the form of electronic communication.

Amplitude modulation, or just AM, is one of the earliest modulation methods that is used in transmitting information over the radio. This technique was devised in the 20th century at a time when Landell de Moura and Reginald Fessenden were conducting experiments using a radiotelephone in the 1900s. After successful attempts, the modulation technique was established and used in electronic communication.

Waveform:**Procedure:**

- Connect the circuit as shown in figure.
- Set the control voltage to 0V by using V_c pot and do not apply modulating signal.
- Observe the output carrier waveform (which is generated by XR2206 itself), and adjust it by using P1 pot.
- Increase the control voltage V_c to 4.5V.
- Set the modulating / message (sine) signal to 1 KHz frequency and adjust the amplitude until perfect AM waveform observed.
- Vary the amplitude of modulating signal to observe under and over modulated AM waveform.

Result: Successfully studied the amplitude modulation and observe the different waveforms.

Teacher's Sign

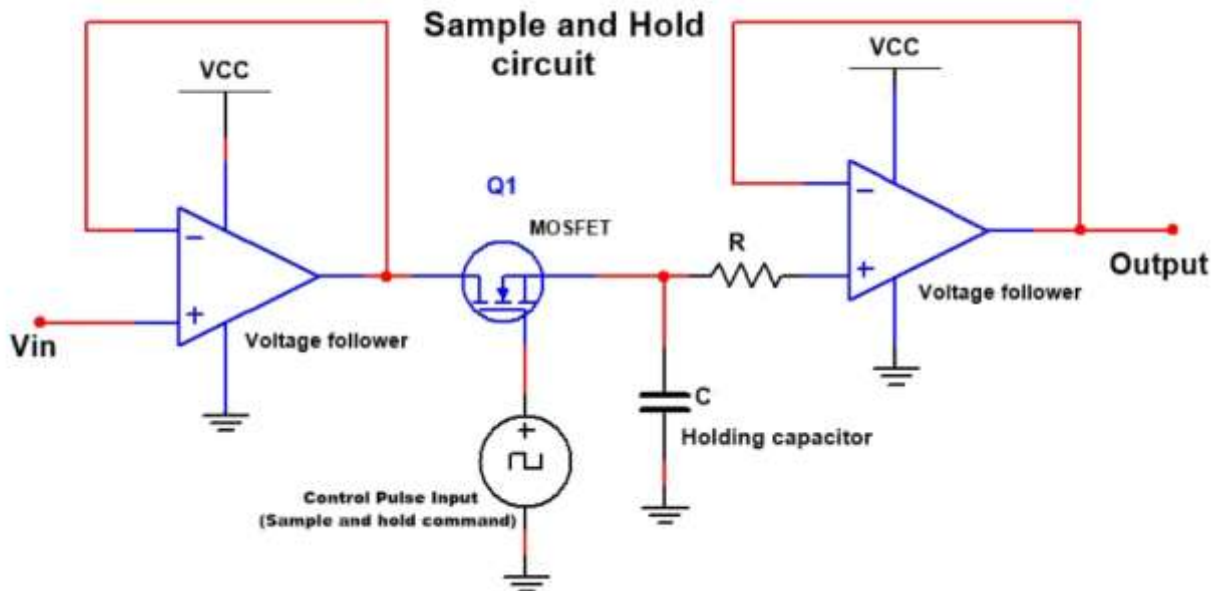


Experiment – 2

Title: Study of Sample and Hold circuit

Aim: To study Sample and Hold circuit and observed the output.

Circuit Diagram:



Theory:

Sample and hold circuit is an electronic circuit that makes the samples of input voltage signals and holds these samples signal for some very small time duration.

A sample and hold circuit is basically an analog to digital converter circuit. Input voltage signals are sampled and held for some duration (in microseconds) using the capacitor, and the output is in the form of a digital pulse. Because this works by the holds the sampled analog input signal, this is called the sample and hold circuit. Holding of the signal is done using a capacitor called a holding capacitor, the capacitor has the ability to store the voltage by charging until it will discharge. This charging and storing property of the capacitor is used here for the holding of the signal.

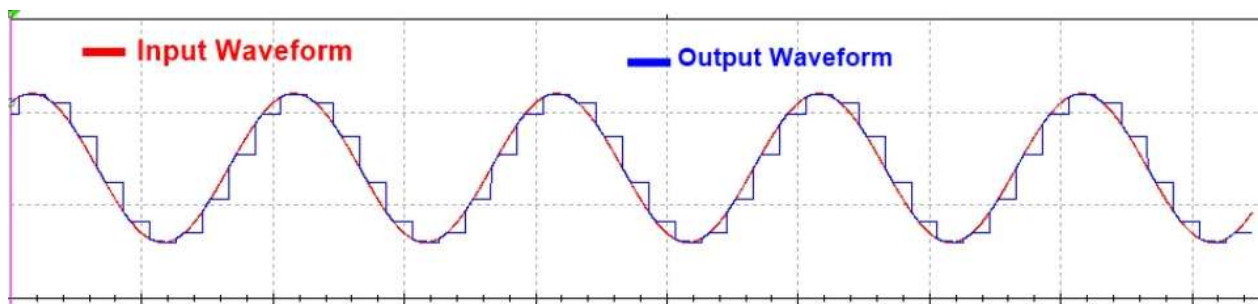
In this circuit, two voltage follower circuits, a capacitor, and A MOSFET are connected. Input voltage given at the control voltage is given on the gate terminal of MOSFET; here, MOSFET is used as a switch. The reason of using the Voltage follower circuit is that this circuit gives output the same as the input signal, and there is a very high input impedance so no load is considered at the input, and the capacitor will also not get discharged because of the very high impedance at the input of the op-amp.

The MOSFET is used here as the control switch for applying the voltage pulse. The Gate terminal of MOSFET is connected with a controlled pulsed signal generator. A stable multivibrator can be used to produce frequency. This astable frequency switch ON-OFF Very fast the MOSFET. By the ON-OFF, ON-OFF of the MOSFET, a pulsed form of the input voltage signal is applied to the NON-inverting

terminal of the Op-amp. The capacitor will charge as a pulse signal gets the 1st pulse to its max level, and this level is HOLD for a definite time by the capacitor because this capacitor will not discharge. Now the next pulse falls, and the capacitor will again charge as this pulse voltage level. and this level holds, and this process is continuously proceeding as per pulse signal and the amplitude level of the pulse signal.

So, in this whole process, the output is sampled and also held for a time duration; these are done by the main capacitor used in this circuit. We know that the sampled signals are the digital signals, so we can say that the sample and hold circuit converts analog or continuous-time signals into digital signals.

Waveform:



Procedure:

- Connect the circuit as shown in figure.
- Set the modulating / Input (Sine) signal to 100 Hz frequency.
- Set the carrier / Pulse signal to 1 KHz frequency.
- Connect any one holding capacitor (either C1 or C2).
- Observe the Output for different values holding capacitor (either C1 or C2). & observe the difference.

Result: Successfully studied the Sample and Hold circuit and observe the different waveforms.

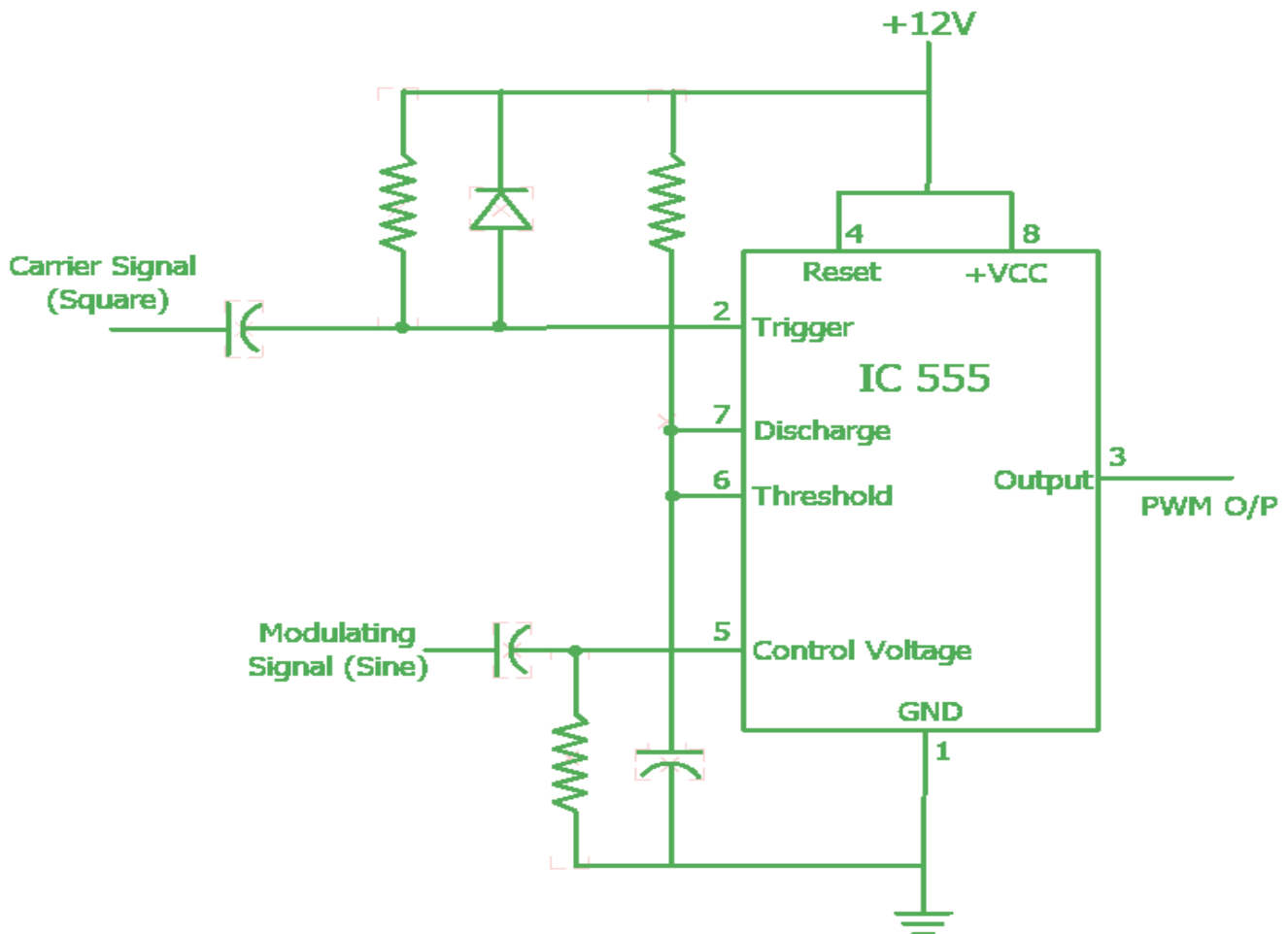
Teacher's Sign

Experiment – 3

Title: Study of Pulse Width Modulation (PWM)

Aim: To study Pulse Width Modulation (PWM) and observed the output.

Circuit Diagram:

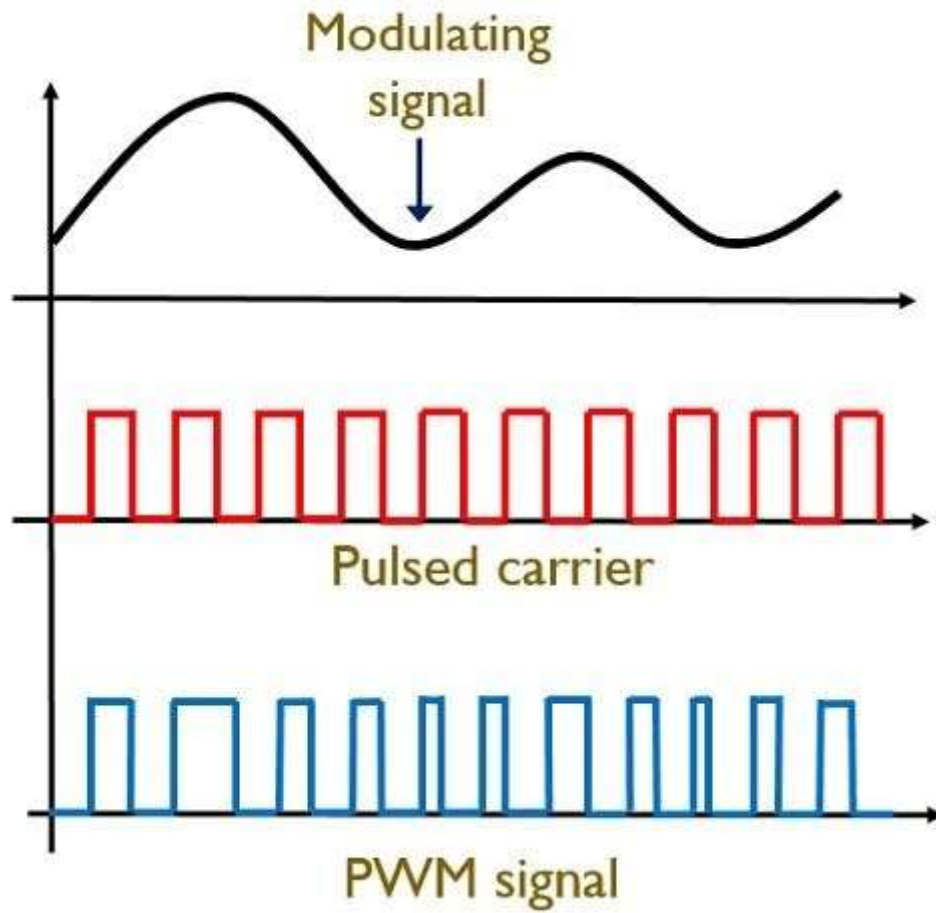


Theory:

In Electronic Engineering, Pulse Width Modulation, or PWM, is a commonly used technique for effectively controlling the power supplied to electrical devices. In order to attain a desired average voltage or power level, the principle of pulse width modulation (PWM) is used for a periodic signal, which is usually a square wave. Pulse-width modulation, commonly known as PWM, is a modulation method that changes the pulse signal's width in electrical systems to regulate the average power supplied to a load. PWM is particularly helpful for effectively regulating the output of audio amplifiers, the speed of motors, and the brightness of light.

A key component of pulse width modulation (PWM) is the duty cycle, which is defined as the ratio of the pulse width to the whole time period. An increase in the duty cycle translates into an increase in average power output.

Waveform:



Procedure:

- Connect the circuit as shown in figure.
- Set the modulating / Input (Sine) signal to 100 Hz frequency.
- Set the carrier / Pulse signal to 1 to 2 KHz frequency.
- Observe the output & trace it.

Result: Successfully studied the Pulse Width Modulation (PWM) circuit and observe the different waveforms.

Teacher's Sign

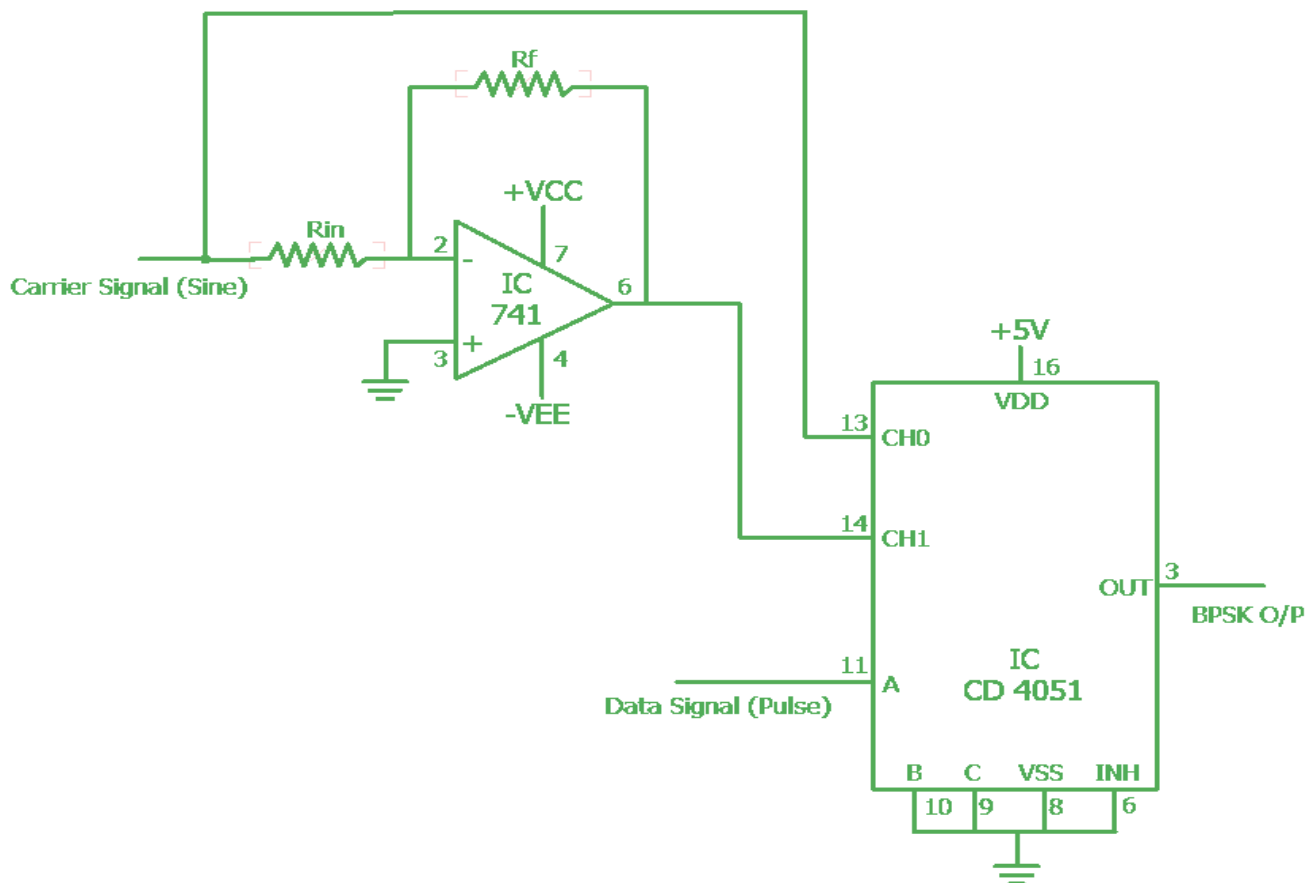


Experiment – 4

Title: Study of Binary Phase Shift Keying (BPSK)

Aim: To study Binary Phase Shift Keying (BPSK) and observed the output.

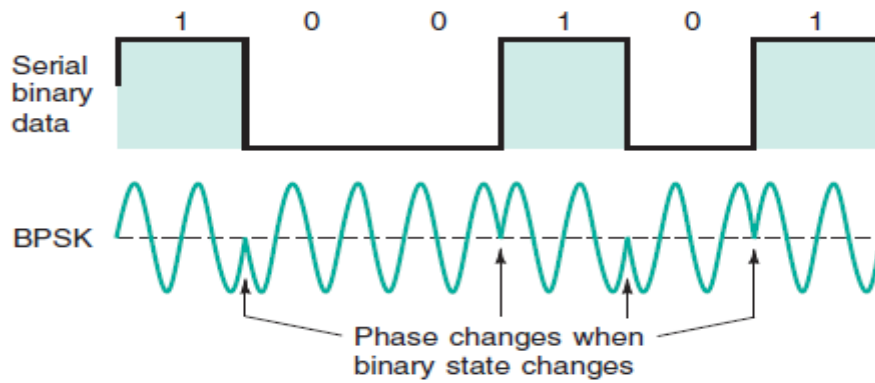
Circuit Diagram:



Theory:

The process of phase-modulating a carrier with binary data is called phase-shift keying (PSK) or binary phase-shift keying (BPSK). The PSK signal uses a 180° phase shift from a reference, but other phase-shift values can be used, for example, 45° , 90° , 135° , or 225° . The important thing to remember is that no frequency variation occurs. The PSK signal has a constant frequency, but the phase of the signal from some reference changes as the binary modulating signal occurs. Binary phase shift keying method has variety of applications in digital communications systems such as the wireless LAN standard, IEEE 802.11, digital modems, wireless telephone networks etc.

In phase-shift keying (PSK), the binary signal to be transmitted changes the phase shift of a sine wave character depending upon whether a binary 0 or binary 1 is to be transmitted. A phase shift of 180° , the maximum phase difference that can occur, is known as a phase reversal, or phase inversion. During the time that a binary 0 occurs, the carrier signal is transmitted with one phase; when a binary 1 occurs, the carrier is transmitted with a 180° phase shift.

Waveform:**Procedure:**

- Connect the circuit as shown in figure.
- Set the modulating / Input (Pulse) signal to 250 Hz frequency.
- Set the carrier / sine signal to 1 KHz frequency by signal generator.
- Adjust the amplitude of sine wave, until proper waveform will be observed.
- Observe the Output & trace it.



Result: Successfully studied the Binary Phase Shift Keying (BPSK) circuit and observe the different waveforms.

Teacher's Sign

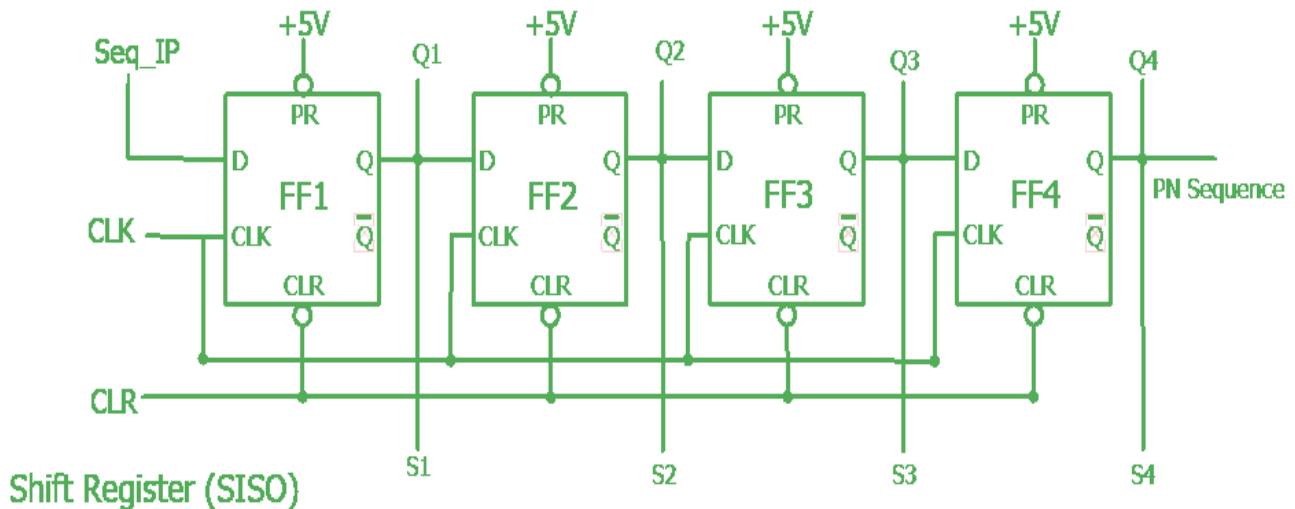
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Experiment – 5

Title: Study of PN Sequence Generator circuit

Aim: To study PN Sequence Generator circuit and observed the output.

Circuit Diagram:



Theory:

Pseudo-Noise (PN) sequences are usually used in order to generate noise that is approximately "white". It has applications in scrambling, cryptography, and spread-spectrum communications. It is also commonly referred to as the Pseudo-Random Binary Sequence (PRBS). These are very widely used in communication standards these days. The qualifier "pseudo" implies that the sequence is not truly random. Actually, it is periodic with a (possibly large) period and exhibits some characteristics of a random white sequence within that period. PN sequences are generated by Linear Feedback Shift Registers (LFSR), as shown in the following Figure.

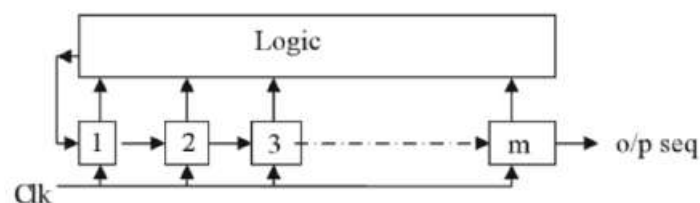


Figure 1: Linear Feedback Shift Registers (LFSR)

A linear feedback shift register (LFSR) is a shift register whose input bit is the output of a linear function of two or more of its previous states (taps). An LFSR of length m consists of m stages numbered $0, 1, \dots, m-1$, each capable of storing one bit, and a clock controlling data exchange. From all the flip-flops of the shift register, outputs are fed to one logic circuit with one switch to each. The output of the logic circuit is again fed to the input of the primary (left most) flip-flop.

For m flip-flop the number of states is 2^m and the maximum length of the period of this sequence is $N = 2^m - 1$. So, for the $m = 4$ the number of states is $2^4 = 16$ and the maximum length of the period is $N = 2^3 - 1 = 15$. Hence, we will observe that after 16^{th} clock pulse we will get the same initialized flip flop content. Therefore, the output sequence depends on the number of flip flops (m), initial state, and feedback logic.

Note that in case that feedback logic consists entirely with modulo-2 (MOD-2) adders (XOR gate) only, the initial state cannot be all zero state because if the state of the shift register is all zero at any time, it remains so for all time and it will not generate new sequences

Procedure:

- Connect the circuit as shown in figure.
- Make CLR switch 1 (make it 0 only when you want to clear output).
- Connect clock input to pulser section.
- Connect "Seq_ip" input signal to Digital i/p.
- Set the initial sequence e.g. S1,S2,S3,S4 -> 0100.
- Then connect the Seq_ip input signal to Ex-OR gate output.
- Ex-OR gate inputs are connected to any of two FF outputs. E.g. Q3 & Q4.
- Apply the clock and note down the output sequence.
- Compare the output sequence with the calculated sequence.

Observation Tabel:

Clock Pulse No.	Intermediate States				Outputs PN Sequence
	S1	S2	S3	S4	
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					
9.					
10.					
11.					
12.					
13.					



14.					
15.					
16.					
17.					



Result: Successfully studied the PN sequence Generator circuit and observe the output.

Teacher's Sign

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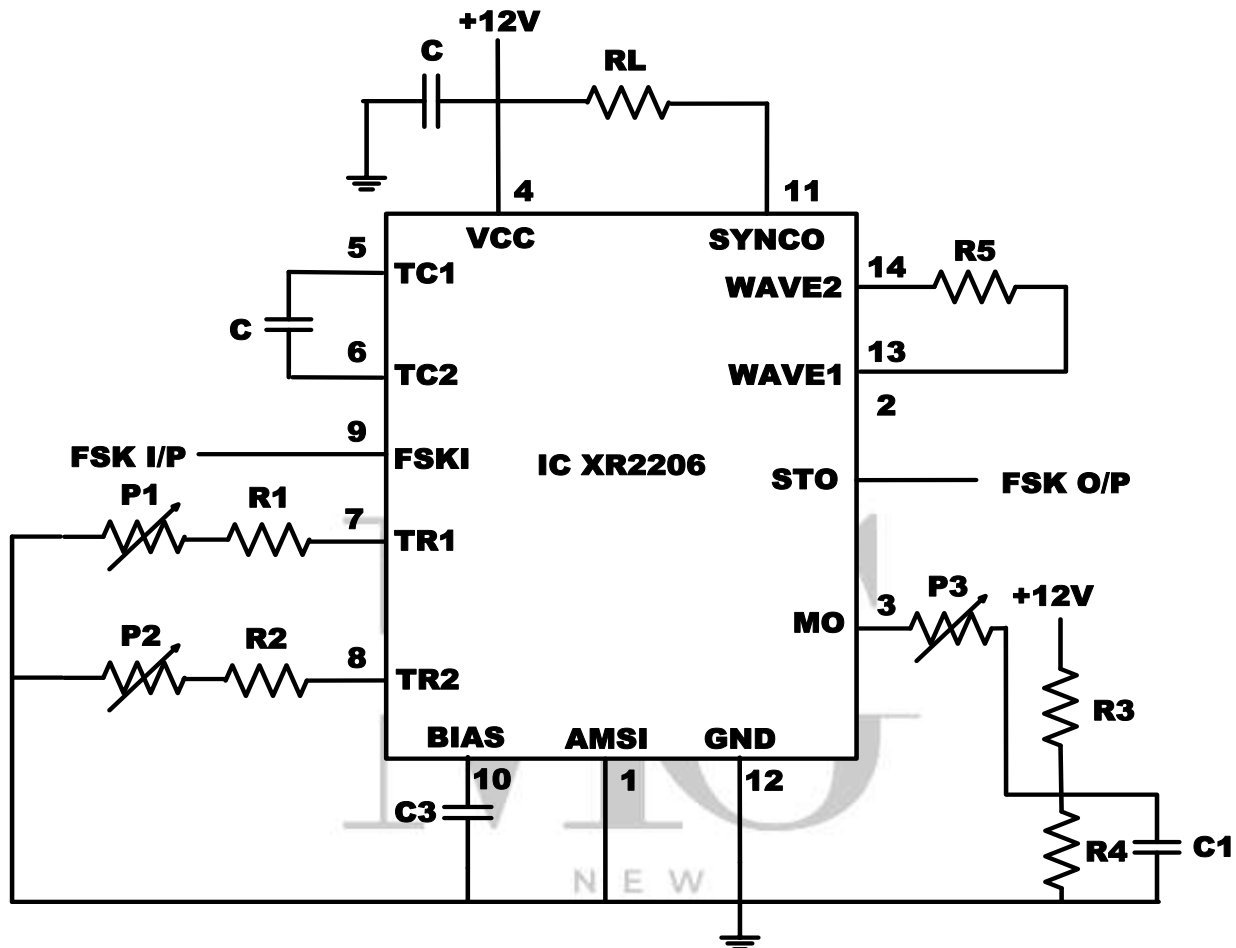


Experiment – 6

Title: Study of Frequency Shift Keying (FSK)

Aim: To study Frequency Shift Keying (FSK) and observed the output.

Circuit Diagram:

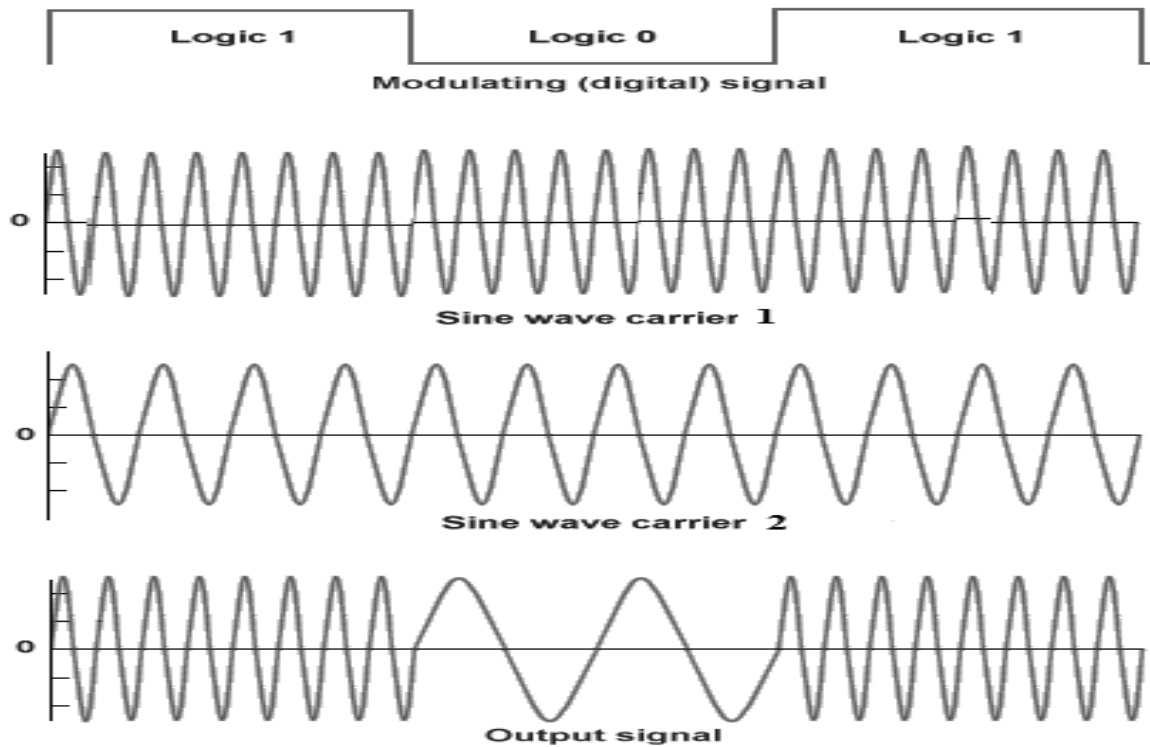


Theory:

Frequency Shift Keying (FSK) is the digital modulation technique in which the frequency of the carrier signal varies according to the digital signal changes. FSK is a scheme of frequency modulation.

The output of a FSK modulated wave is high in frequency for a binary High input and is low in frequency for a binary Low input. The binary 1s and 0s are called Mark and Space frequencies. The technology is used for communication systems such as telemetry, weather balloon radiosondes, caller ID, garage door openers, and low frequency radio transmission in the VLF and ELF bands. The simplest FSK is binary FSK (BFSK), in which the carrier is shifted between two discrete frequencies to transmit binary (0s and 1s) information. According to Bell 103 standard, there are two frequencies assigned for data as follows

Mark frequency $\rightarrow 1270$ Hz and Space frequency $\rightarrow 1070$ Hz

Waveform:**Procedure:**

- Connect the circuit as shown in figure.
- Apply zero volts (Logic L) to the input of FSK generator.
- Adjust P2 to adjust Space frequency.
- Apply +5 V (Logic H) to the input of FSK generator.
- Adjust P1 to adjust Mark frequency.
- After performing calibration of output frequencies, connect FSK input signal either to 0V or 5V and note down the readings of Mark and Space frequencies.
- Then apply TTL square wave input signal to the FSK input, observe the output waveform and trace it.

FSK Output Observations -

Type of Signal	Given (Set) Frequency	Observed Frequency
Sine wave for TTL logic level '1' (Mark signal)		
Sine wave for TTL logic level '0' (Space signal)		



Result: Successfully studied the Frequency Shift Keying (FSK) circuit and observe the different waveforms.

Teacher's Sign

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Experiment – 7

Title: Study of Time Division Multiplexing (TDM)

Aim: To study Time Division Multiplexing (TDM) and observed the output.

Circuit Diagram:

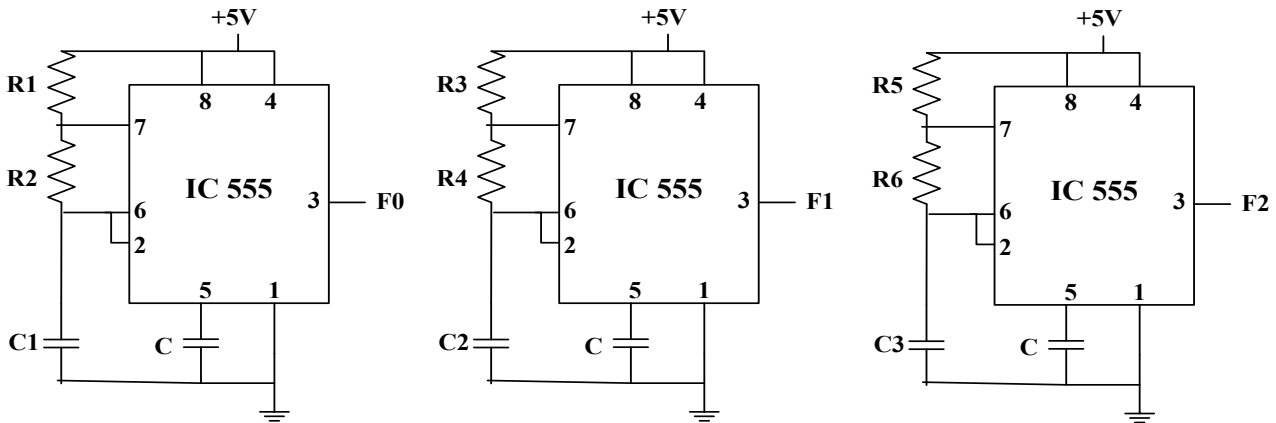


Figure 1 - F0, F1 & F2 frequency generator

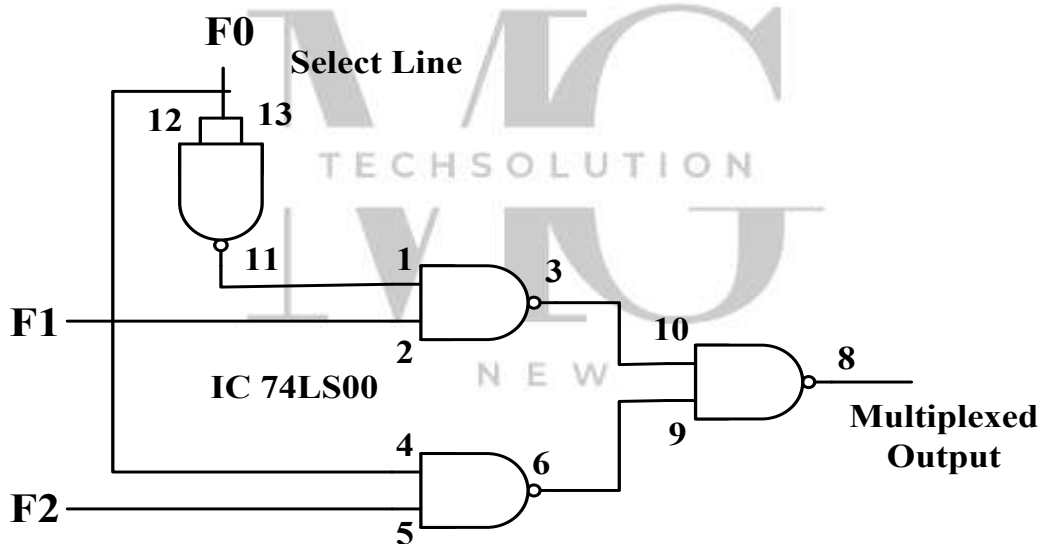


Figure 2 - Multiplexer Circuit

Procedure:

- Connect the circuit for individual astable multivibrator as shown in the figure 1.
- See the outputs on CRO and measure the frequency of the signal waveform F1, F2 & Select line frequency F0.
- Calculate frequency using the formula:

$$F = (1.44) / (R1 + 2R2) C$$

Where, C1, R1 and R2 values are obtained from the circuit panel.
- Verify the calculated frequency with the observed frequency reading.

- Connect outputs of the Astable Multivibrators (F0, F1, F2) to the multiplexer circuit (obtained using NAND gates) shown in the figure 2.
- See the output waveform of multiplexer on CRO and understand multiplexing action.

Observation table:

1. Measurement of Frequency on C.R.O.

Signal	Calculated	Observed
F0		
F1		
F2		

2. Multiplexer output:

Sr. No.	Logic Level of select Signal F0 (Manually)	Observed Frequency at Multiplexer Output
1.	0	
2.	1	



Result: Successfully studied the Time Division Multiplexing (TDM) circuit and observe the different waveforms.

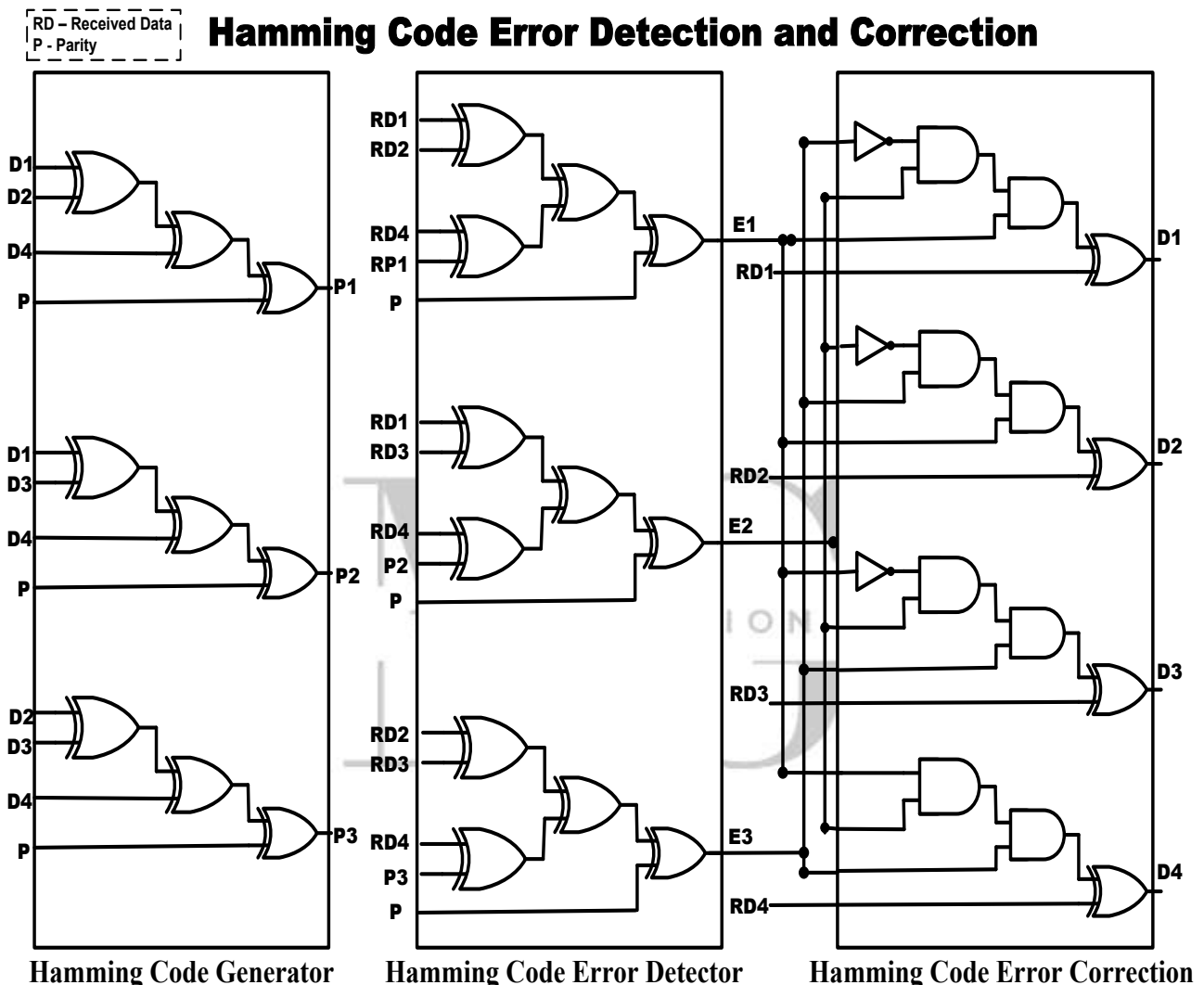
Teacher's Sign

Experiment – 8

Title: Error Detection and Correction using Hamming code

Aim: Study of Error detection and correction by using Hamming Code technique.

Circuit Diagram:



Theory:

In coding theory, Hamming is a linear error-correcting code, refers to specific code that Richard W. Hamming introduced in 1950. At the time, Hamming worked at Bell Telephone Laboratories and was frustrated with the error-prone punched card reader, which is why he started working on error-correcting codes. Hamming code is one of the single bit error detection and correction methods. When transmission of digital signals takes place between two systems such as in between two computers the noise can introduce an error in the binary bits. Error means 0 change to 1 and 1 may be change 0 and transmitted data corrupted due to error.

Hamming code is nothing but one or more than one parity bits inserted in a typical manner to detect the error.

1	2	3	4	5	6	7
001	010	011	100	101	110	111
P1	P2	D1	P3	D2	D3	D4

Table 1: Arrangement of Hamming Code

Where, D= data bits, P = Parity bits.

The parity bits are inserted at each 2^n bit where $n=0, 1, 2, 3, \dots$

So P1 is at $2^0 = 1$ & P2 is at $2^1 = 2, \dots$ etc.

There is a simple mathematical equation to find parity bits.

$$2^k \geq n+k+1 \text{ where}$$

k = number of parity bit, n = number of data bits

If there are four data bits, we have to find no. of parity bits,

For $n=4$, consider $k=1$, $2^1 \geq 4+1+1$

$2 \geq 6$ which is not true hence condition is not satisfied.

For $n=4$, $k=2$, $2^2 \geq 4+2+1$

$4 \geq 7$ which is not true hence condition is not satisfied.

For $n=4$, $k=3$, $2^3 \geq 4+3+1$

$8 \geq 8$ which is true hence condition is satisfied.

Hence no. of parity bits are equal to 3. Hence total no. of bits in hamming code are data bits plus parity bits, which is equal to 7.

For finding a parity bits there is need to be checked particular bits in data (here we consider data bits equal to 4, hence parity bits are 3), which is shown in the following table (refer to table 1):

Parity Bit	Bits to be checked
P1	3, 5, 7
P2	3, 6, 7
P3	5, 6, 7

Problems:-

Q: A bit word 1011 is to be transmitted. Construct the even parity seven-bit Hamming code for this data.

P1	P2	D1	P3	D2	D3	D4
		1		0	1	1

The parity bit P1 checks the $P1 = \{3, 5, 7\} = \{D1, D2, D4\} = \{1, 0, 1\}$

The location for P1 had the TWO numbers of 1 therefore we has to set $P1 = 0$ for even parity

Similarly for $P2 = \{3, 6, 7\} = \{D1, D3, D4\} = \{1, 1, 1\}$

The location for P2 had THREE number of 1 therefore we has to set $P2=1$ for even parity.

Similarly for $P3 = \{5, 6, 7\} = \{D2, D3, D4\} = \{0, 1, 1\}$

The location for P3 had the two numbers of 1 therefore we have even parity only. Therefore $P3 = 0$.



The hamming code for transmitting is

P1	P2	D1	P3	D2	D3	D4
0	1	1	0	0	1	1

Q: If the 7-bit Hamming code word received by a receiver is 1010011. Assuming the even parity state whether the received code word is correct or wrong. If wrong, locate the bit in error.

Ans.: - For 3 parity bit hamming code there are 3 error bits corresponding to each parity bit.

The received code word

P1	P2	D1	P3	D2	D3	D4
1	0	1	0	0	1	1

Step 1:- Check the bits P1 D1, D2 & D4

$$E1 = \{P1, D1, D2, D4\} = \{1, 1, 0, 1\} = \text{odd parity}$$

Therefore error exists here.

Therefore $E1 = 1$ in the 1st position of the error word.

Step 2:- check the bits P2, D1, D3 & D4

$$E2 = \{P2, D1, D3, D4\} = \{0, 1, 1, 1\} = \text{odd parity}$$

Therefore error exists here.

Therefore $E2 = 1$ in the 2nd position of error word.

Step 3:- check the bits P3, D2, D3 & D4

$$E3 = \{P3, D2, D3, D4\} = \{0, 0, 1, 1\} = \text{even parity}$$

Therefore no error here

Therefore $E3 = 0$ in the 3rd position of the error word.

Step 4:- Error word

$$E = \{E3, E2, E1\} = \{0, 1, 1\} = (3)_{10}$$

Therefore 3rd bit position of the transmitted code word is in error

Therefore 1 replaced with 0.

Therefore the correct data is

P1	P2	D1	P3	D2	D3	D4
1	0	0	0	0	1	1

Limitation:- The main limitation or disadvantage of hamming code is that it is an only single bit error detector method. It can't find the more than one bit error.



Truth Table: (Hamming code generator for Even Parity)

Data Bits				Parity Bits		
(D1)	(D2)	(D3)	(D4)	P1	P2	P3

Truth Table: (Hamming code generator for Odd Parity)

Data Bits				Parity Bits		
(D1)	(D2)	(D3)	(D4)	P1	P2	P3

Truth Table: (Hamming code error detection and correction)

Transmitted Hamming Code							Received Hamming Code							Error Bits			Corrected Data			
P1	P2	D1	P3	D2	D3	D4	P1	P2	D1	P3	D2	D3	D4	E1	E2	E3	D1	D2	D3	D4

Procedure:

- Connect the power cord of the trainer kit to the AC mains supply and turn it ON.
- Firstly select the same parity for hamming code generator and error detector circuit (either even or odd).

- Select the data bits D1-D4 and Parity bit (0 – Even, 1 - Odd) using toggle switches.
- Note down the 7 bit hamming code.
- In first step do not add error in the transmitted hamming code and note down the received hamming code, and error bits. For no error condition, error bits should be zero.
- In next step, add single bit error in the transmitted hamming code and note down the received data, error bits and corrected data bits. Error bits will show the position of the error.



Result: The hamming code generation, error detection and correction technique is studied successfully.

Teacher's Sign.

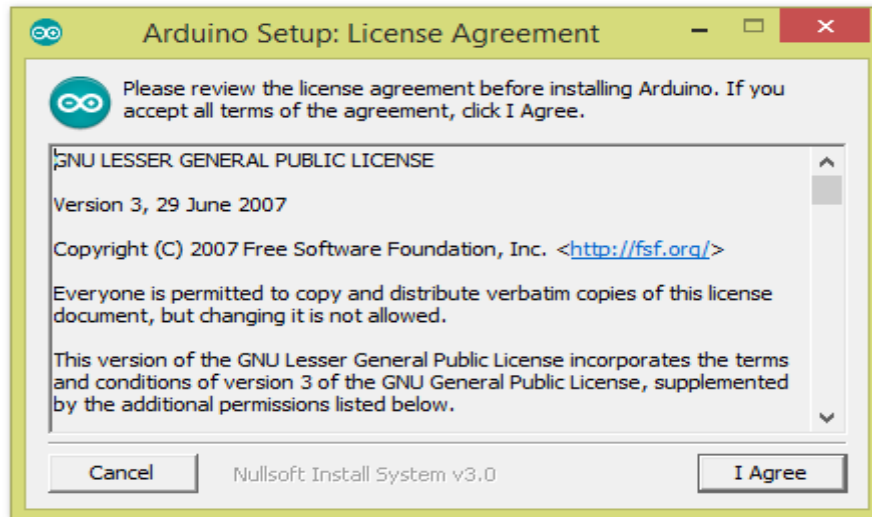


User Guide for Arduino IDE Software

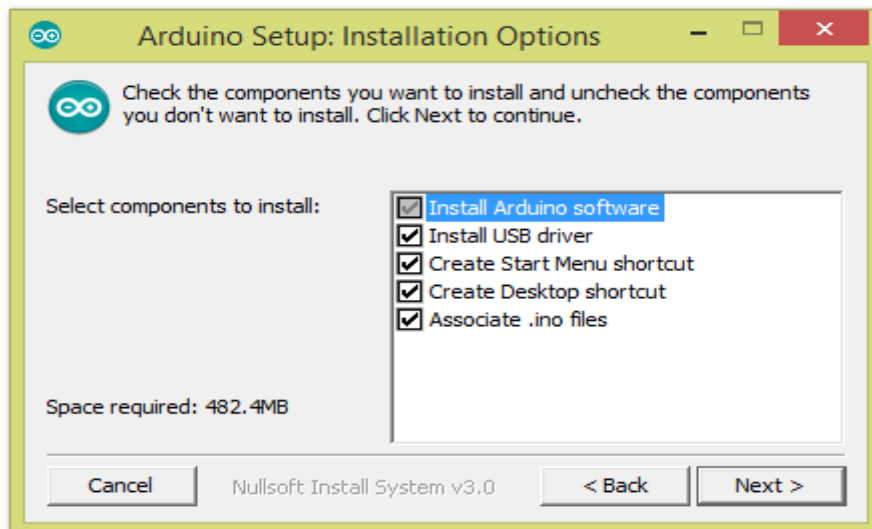
Steps to Installation of Arduino IDE software

Download Link: <https://www.arduino.cc/en/Main/Software>

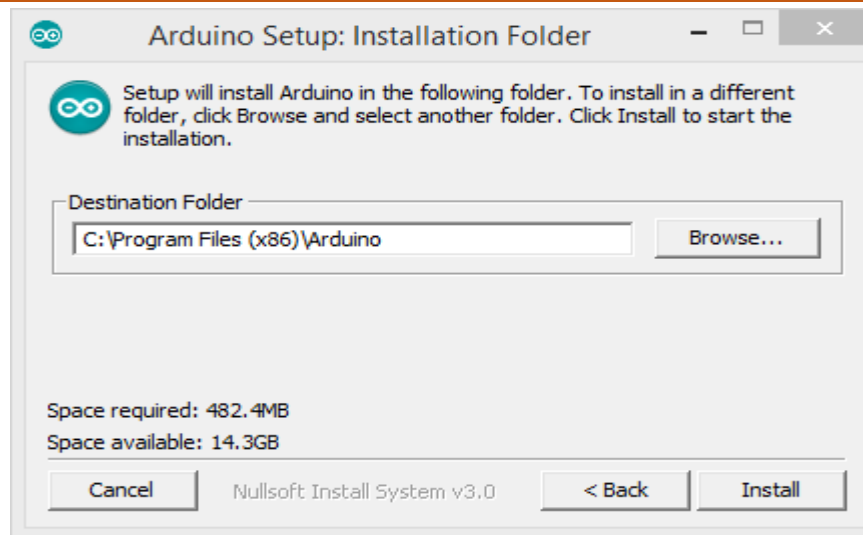
1. Double Click on arduino.exe file.
2. Then click on “I Agree”.



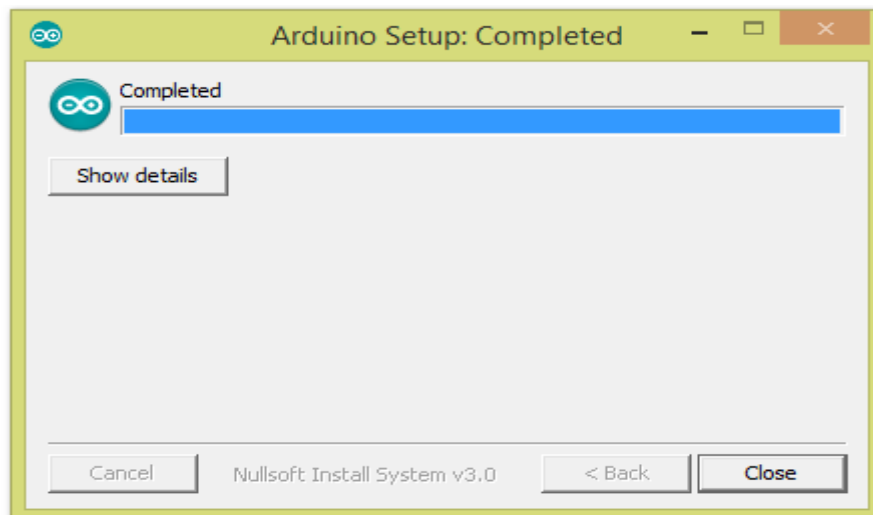
3. Without changing anything click on “Next >”.



4. Select the destination folder & click on “Install”.

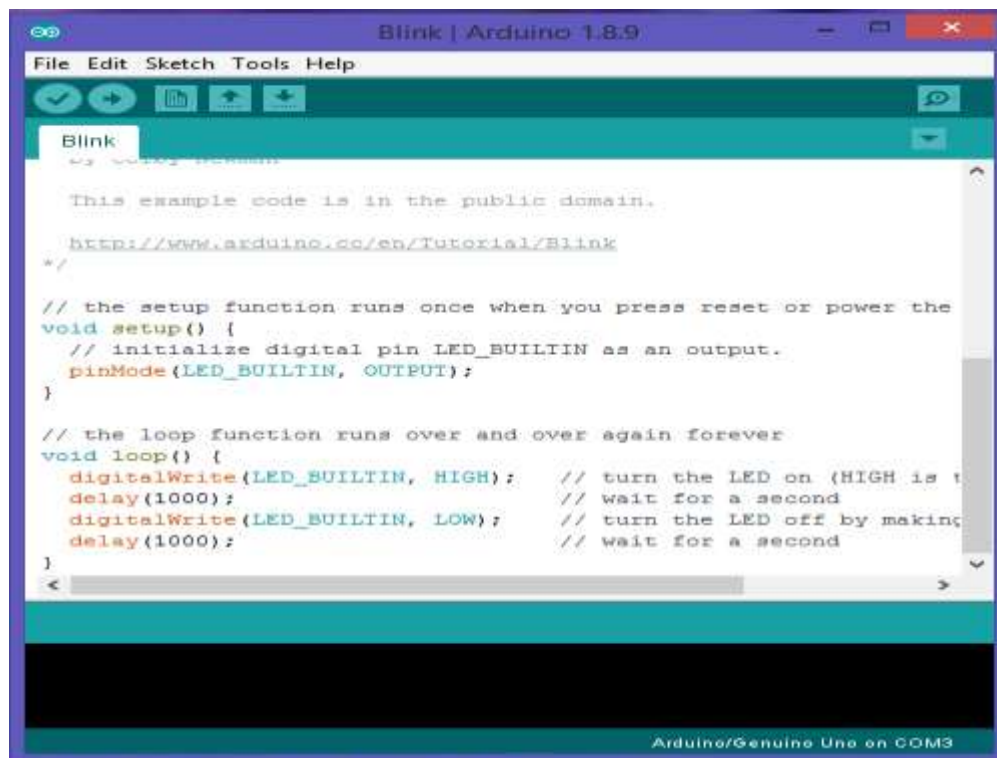


5. After installation completed click on “Close”.

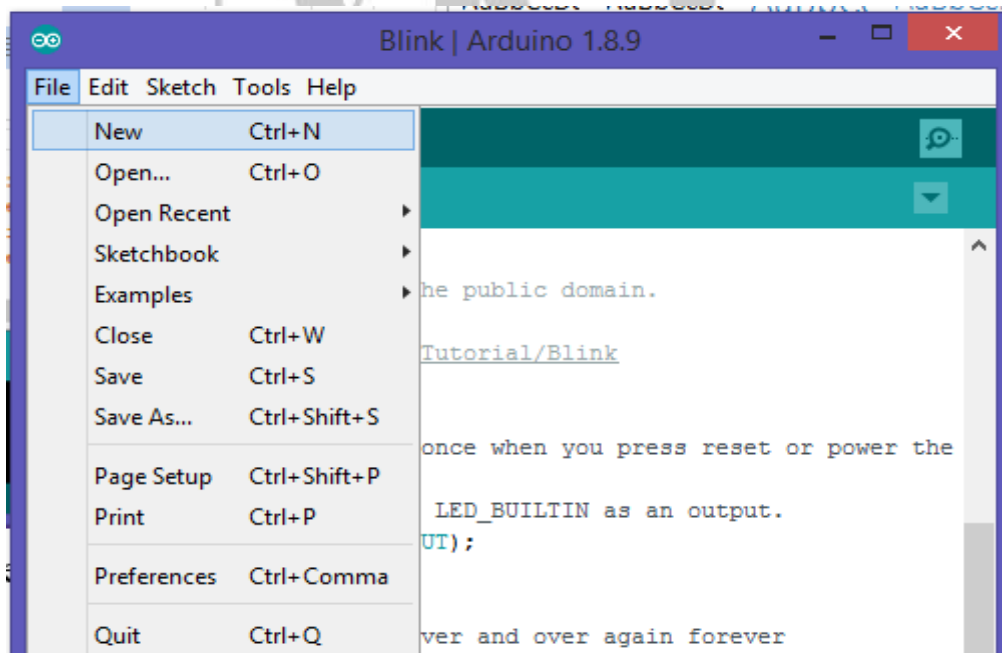


Steps to use of Arduino IDE software

1. Double Click on “Arduino” icon.
2. The Arduino IDE will be opened with previously open project.



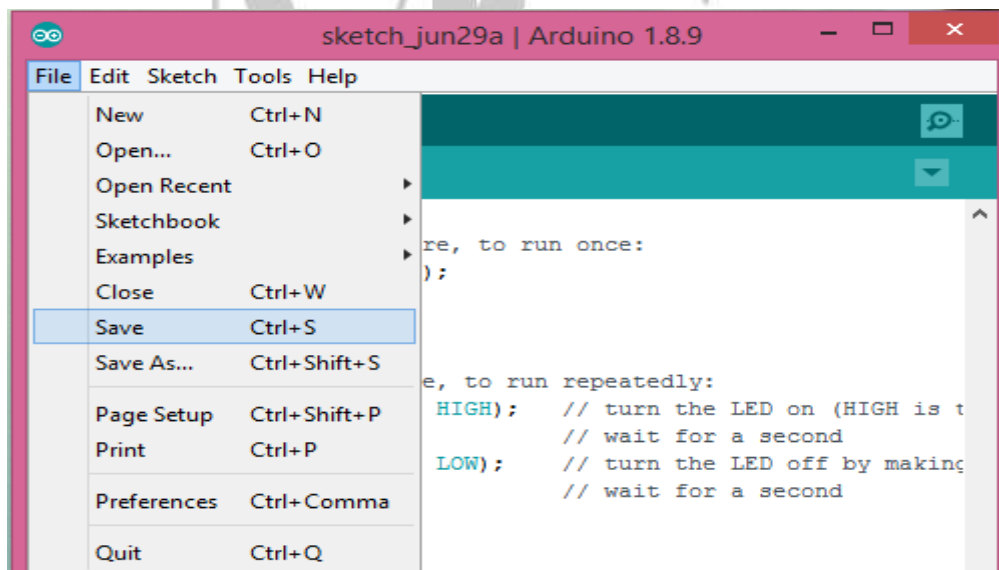
3. For new project go to File -> New.



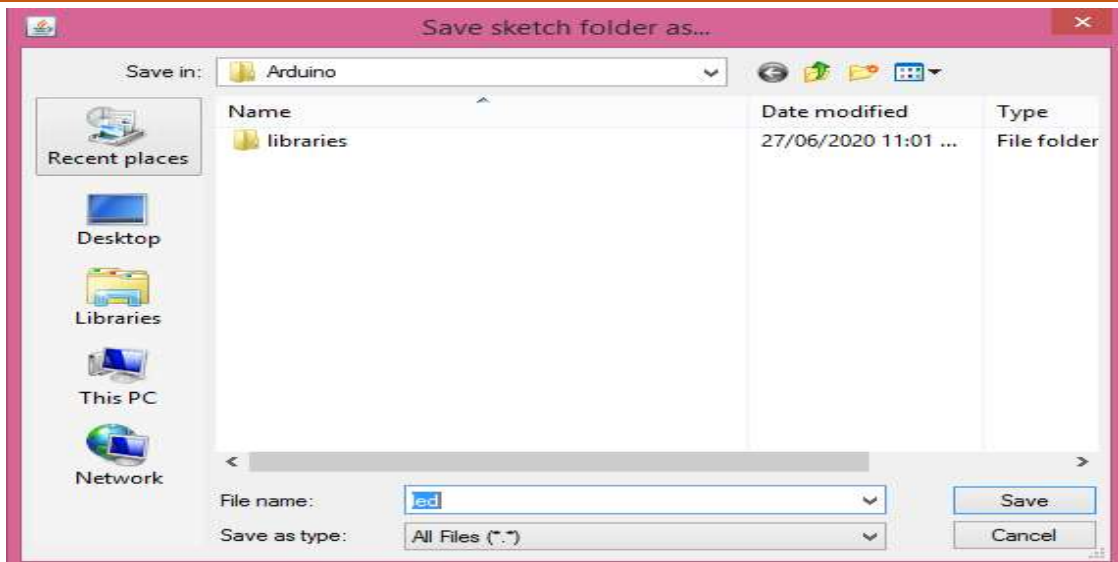
4. New Project is open.
5. It contains two main sections. setup() & loop().



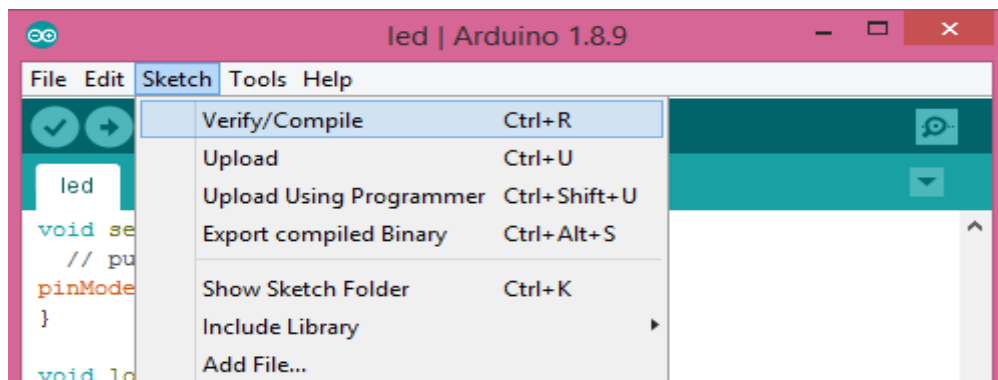
6. After writing your program save it, go to File -> Save.



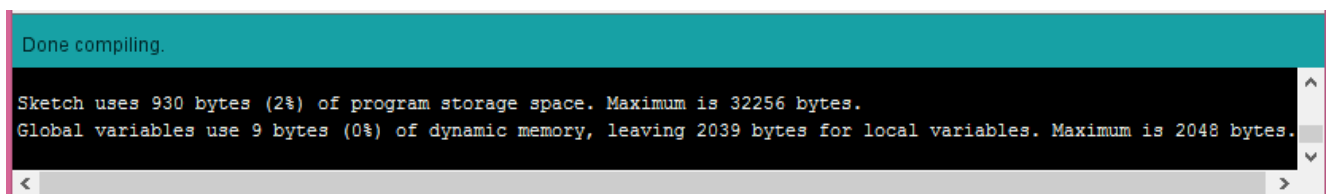
7. Select the folder where you want to save your file (if don't want to save in default Arduino folder), then give the appropriate name to the file and click on "save".



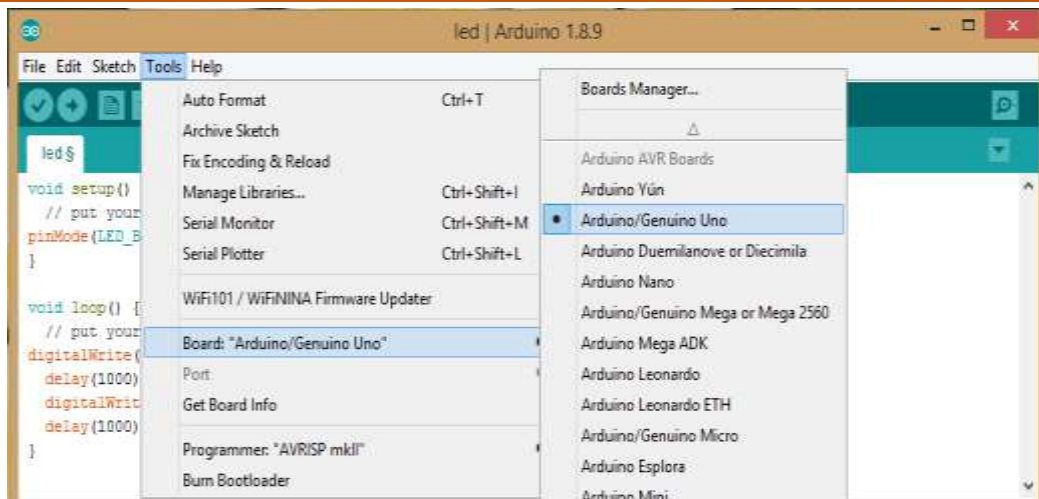
8. Then to compile the written program, select Sketch -> Verify / Compile option. Or click on “Verify” button.



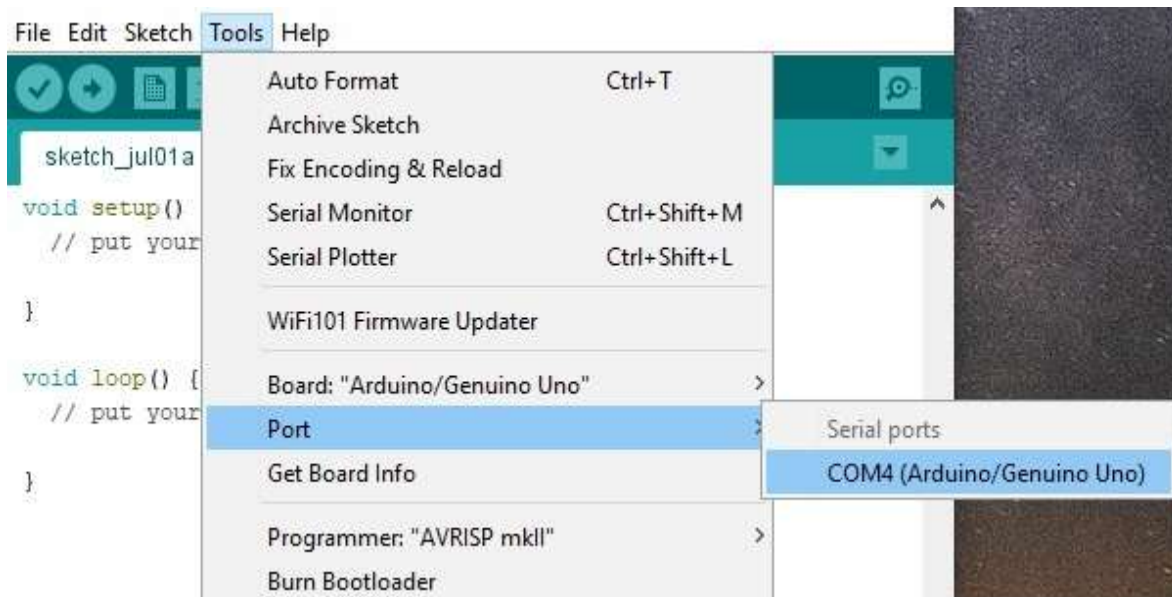
9. After compilation is done, output window display message “DoneCompiling” (if there are no errors in a program). Also displays information about how much program & data memory is used by the program.



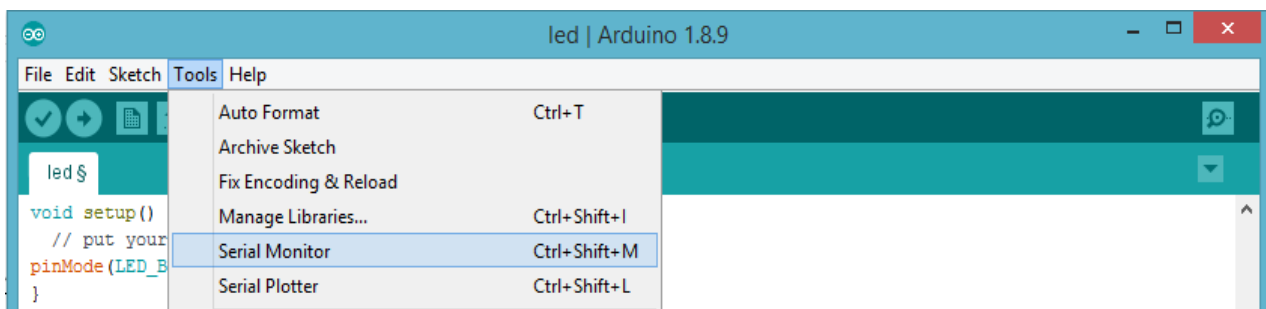
10. Then select the desired Arduino board, go to Tools -> Board: -> Arduino / Genuino Uno (if Uno is used) or select Arduino Nano (if nano is used).



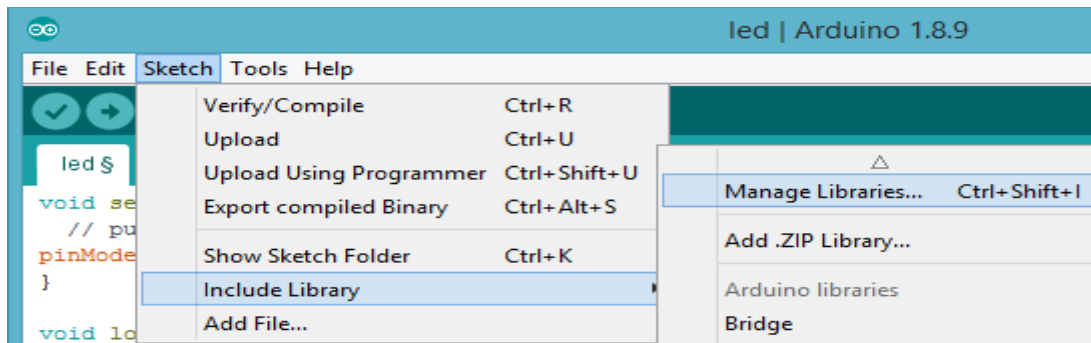
11. After selecting the board, select the COM port where selected board is connected. Go to the Tools -> Port.



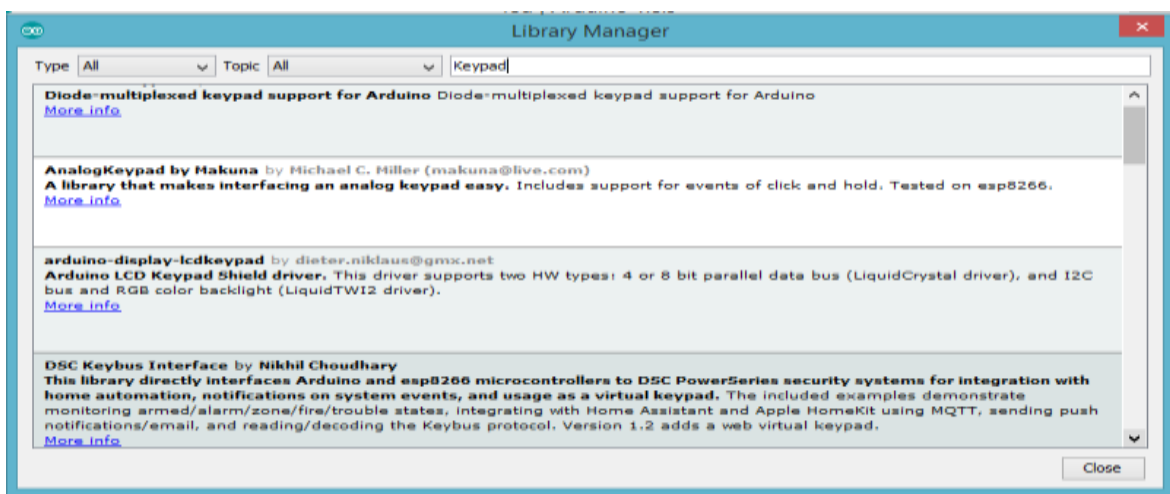
12. Once board & COM port is selected, upload the program on to the connected board. Go to Sketch -> Upload. (Arduino has on board programmer, hence no need of external programmer.)
13. Perform necessary connections to the Arduino and observed the output. In order to execute program again from start press Reset button on the Arduino board.
14. If you want to use serial monitor, go to Tools -> Serial Monitor. (Serial monitor displays the data coming from serial port of the Arduino).



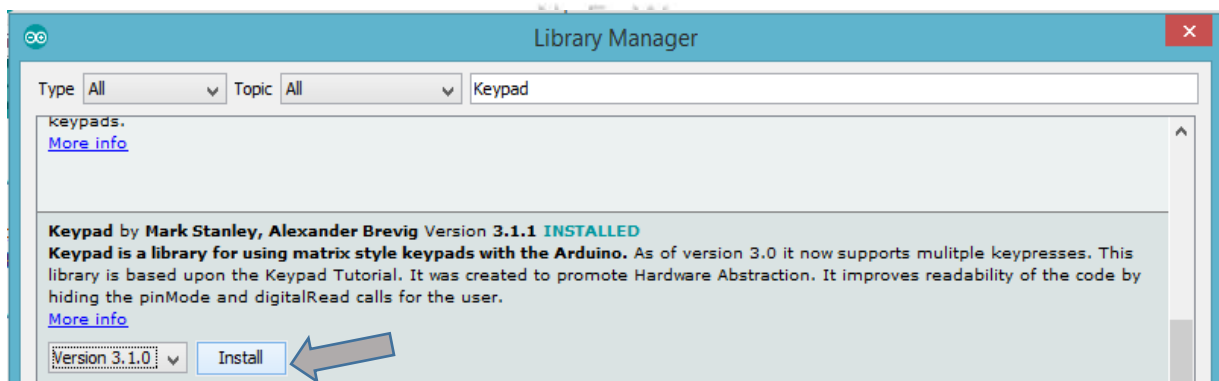
15. If library for any component (sensor / shield) is not available in the Arduino IDE, it can be added externally. Go to Sketch -> Include Library -> Manage Libraries... (Requires internet connection)



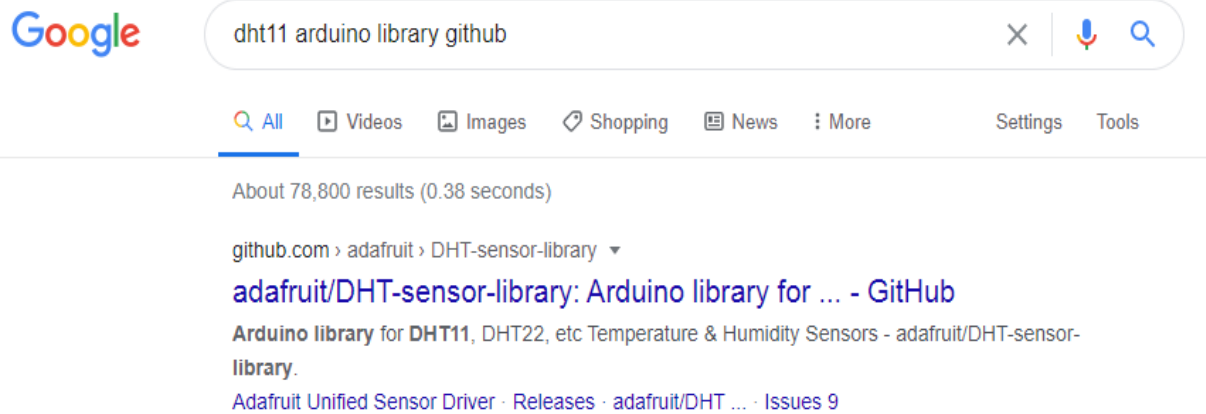
16. The new window will open, where you can search the library you want. E.g. type the name keypad for keypad library, as follows:



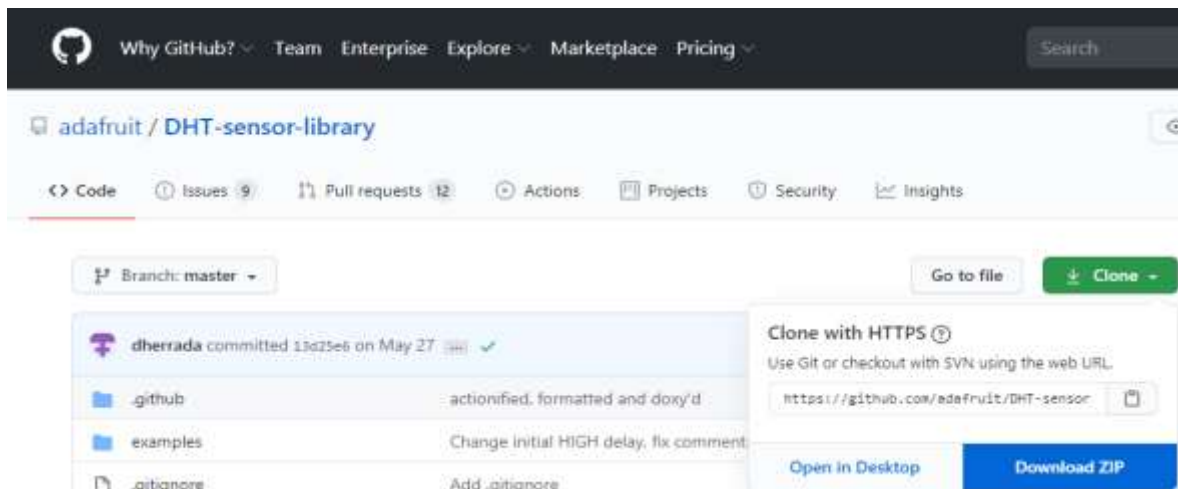
17. Select the desired Keypad library, select version and click on install.



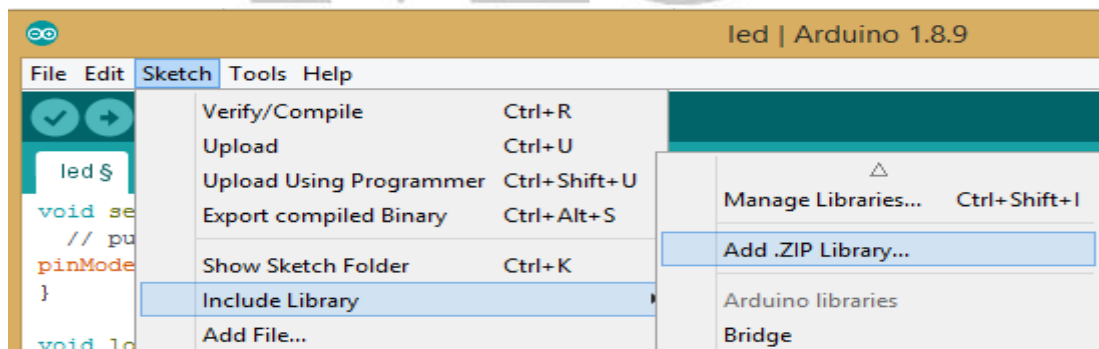
18. If the desired library is not found here, then there is another option to add library. Search the library in the GitHub. E.g. search for DHT11 sensor library.



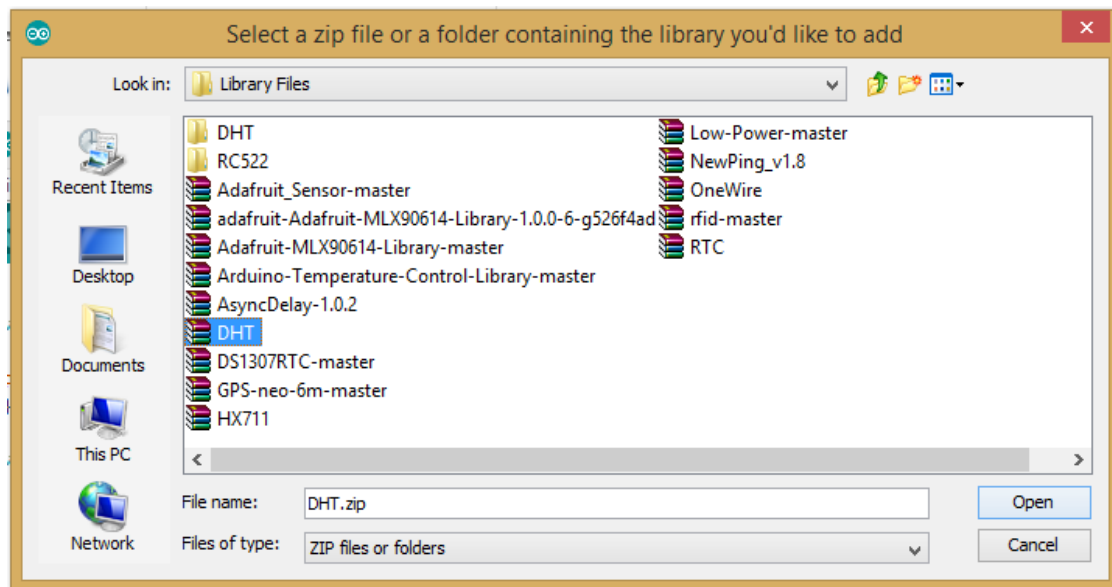
19. In the GitHub website select “Download ZIP” option.



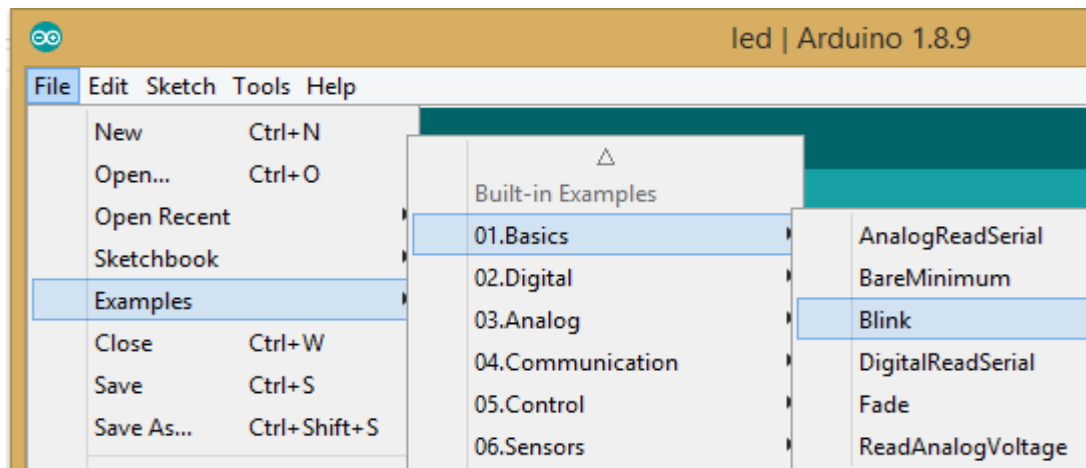
20. Then go to Arduino IDE, select Sketch -> Include Library -> Add .ZIP Library...



21. Select the downloaded ZIP file and click on “Open”.



22. You can go through the in-build examples available in the Arduino IDE and try them on actual Arduino board as per your interest.



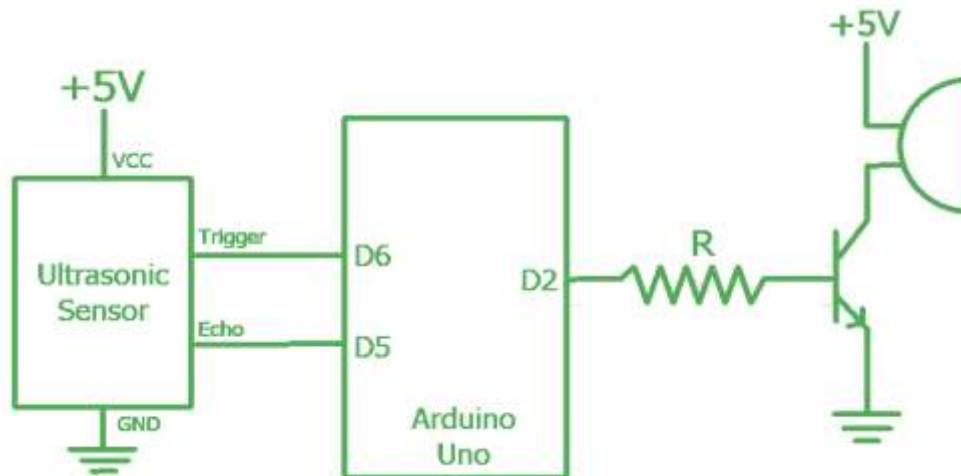
Experiment – 9

Title: Ultrasonic sensor for motion detection using Arduino

Aim: To study interfacing of ultrasonic sensor for motion detection using Arduino and write a program for the same.

Software Used: Arduino IDE

Circuit Diagram:



Theory:

An Ultrasonic Sensor is a device that measures distance to an object using Sound Waves. It works by sending out a sound wave at ultrasonic frequency and waits for it to bounce back from the object. Then, the time delay between transmission of sound and receiving of the sound is used to calculate the distance. It is done using the formula $\text{Distance} = (\text{Speed of sound} * \text{Time delay}) / 2$. We divide the distance formula by 2 because the sound waves travel a round trip i.e from the sensor and back to the sensor which doubles the actual distance.

The HC-SR04 is a typical ultrasonic sensor which is used in many projects such as obstacle detector and electronic distance measurement tapes.

Procedure:

- Connect the USB cable between Arduino & computer (PC).
- For interfacing ultrasonic sensor to Arduino, connect D6 pin of Arduino to I/O1 & D5 pin to I/O2 pin.
- Then connect the 4 pin connector of Arduino section to 4 pin connector of ultrasonic sensor.
- If you want to see the sensor data on serial monitor, open the serial monitor of Arduino IDE.
- If you want to see the sensor data on LCD, connect the second 4 pin connector of Arduino section to LCD section. Then connect the A4 pin of Arduino to I/O3 & A5 pin of Arduino to I/O4.
- Observe the output.

Program:**Using Serial Communication -**

```
#include <NewPing.h>
#include <SoftwareSerial.h>
#define TRIGGER_PIN 6
#define ECHO_PIN 5
#define MAX_DISTANCE 200
NewPing sonar(TRIGGER_PIN, ECHO_PIN, MAX_DISTANCE);

void setup() {
  Serial.begin(9600);
  pinMode(2, OUTPUT);
  delay(1000);
}

void loop() {
  delay(500);
  int distance = sonar.ping_cm();
  Serial.println(distance);
  if (distance <= 05)
  {
    digitalWrite(2, HIGH);
  }
  else
  {
    digitalWrite(2, LOW);
  }
}
```

Result: Interfacing of ultrasonic sensor for motion detection is successfully studied, tested and observe the output.

Teacher's Sign.

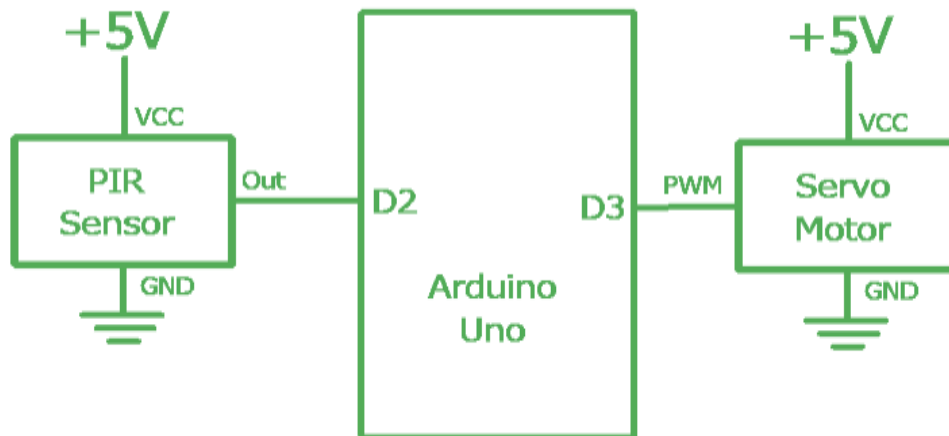


Experiment – 10

Title: Arduino based Automatic Door Control System using Servomotor

Aim: To study interfacing of servo motor and PIR sensor for automatic door control system application.

Circuit Diagram:



Theory:

A passive infrared sensor (PIR) is an electronic sensor that measures infrared light radiating from objects in its field of view. They are most often used in PIR-based motion detectors. PIR sensors allow you to sense motion. They are small, inexpensive, low-power, easy to use and don't wear out. Hence are commonly used in security alarms and automatic lighting applications. Generally, PIR sensor can detect animal/human movement in a requirement range. PIR is made of a pyroelectric sensor, which is able to detect different levels of infrared radiation. The detector itself does not emit any energy but passively receives it.

A servo motor is a precision motor that uses a servo motor working principle based on closed-loop feedback control, which can rotate to specific angles with exceptional accuracy. If you want to rotate an object to specific angles or distances, servo motors are the ideal choice due to their servo mechanism with built-in position feedback. Popular servo motor models include SG90 (micro servo), MG995 (metal gear servo) etc.

Procedure:

- Connect the USB cable between Arduino & computer (PC).
- For interfacing PIR sensor & servo motor to Arduino, connect D2 pin of Arduino to I/O1 & D3 pin to I/O2 pin.
- Then connect the 4 pin connector of Arduino section to 4 pin connector of automatic door control system section.
- If you want to see the sensor data on serial monitor, open the serial monitor of Arduino IDE.

- If you want to see the sensor data on LCD, connect the second 4 pin connector of Arduino section to LCD section. Then connect the A4 pin of Arduino to I/O3 & A5 pin of Arduino to I/O4 pins.
- Observe the output.

Program:

```
#include <Servo.h>

Servo myservo1;

const int sensorPin = 2;    // PIR sensor pin
const int servoPin = 3;    // servo pin
int sensorState = 0;

void setup() {
  pinMode(servoPin, OUTPUT);
  myservo1.attach(servoPin);
  pinMode(sensorPin, INPUT);
  myservo1.write(0);
  delay(2000);
}

void loop() {
  sensorState = digitalRead(sensorPin);
  if (sensorState == HIGH) {
    myservo1.write(90);    // Door Open:
    delay(4000);
  } else {
    myservo1.write(0);    // Door Close:
  }
  delay(1000);
}
```

Result: Study of interfacing servo motor and PIR sensor for automatic door control system application is successfully studied.

Teacher's Sign.

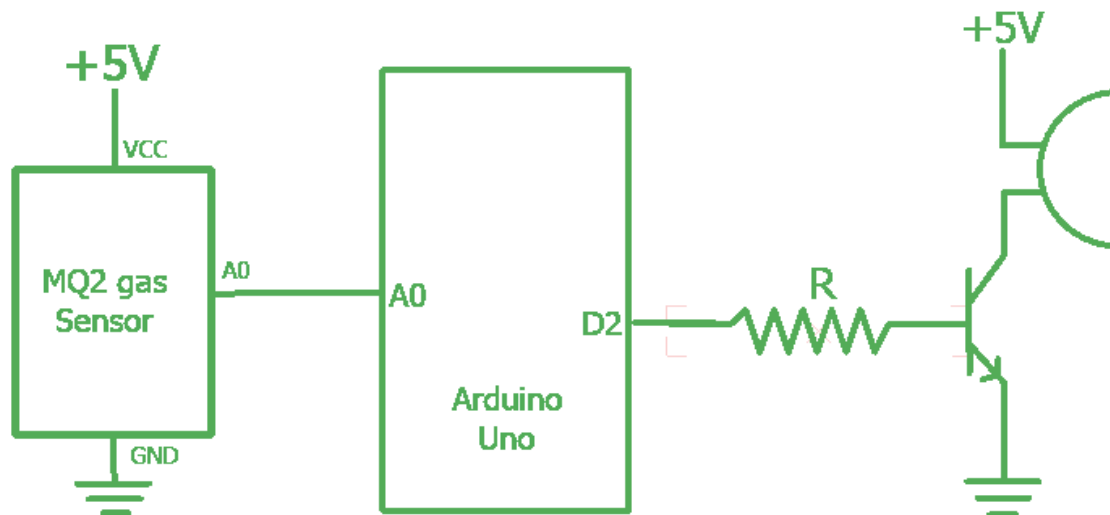


Experiment – 11

Title: Gas Leak Detection and controlling using gas sensor using Arduino

Aim: To study interfacing of Gas sensor using Arduino.

Circuit Diagram:



Theory:

The MQ2 is a Metal Oxide Semiconductor (MOS) gas sensor used to detect various gases, including smoke, propane, LPG, and carbon monoxide, by measuring changes in electrical resistance. It is known as a chemi resistor because its resistance decreases as the concentration of gases around it increases. The sensor can provide both an analog voltage output proportional to the gas concentration and a digital output that can be adjusted with a potentiometer.

Key features -

Gas detection: Sensitive to a wide range of combustible gases such as LPG, propane, methane, hydrogen, alcohol, smoke, and carbon monoxide.

Working principle: A tin oxide sensing material changes its conductivity when it comes into contact with flammable gases. The higher the gas concentration, the higher the conductivity (lower resistance).

Output: It provides a 0V to 5V analog output signal that corresponds to the gas concentration. It also has a digital output that can be set to trigger at a specific gas level using the onboard potentiometer.

Detection range: Can detect gas concentrations in the range of approximately 200 to 10,000 parts per million (ppm).

Power and operation: Typically runs on a 5V DC power supply and requires a preheating time of about 20 seconds before it can be used for accurate measurements.

Applications: Commonly used for gas leakage detection in homes and industries, and in various safety and monitoring projects.



Procedure:

- Connect the USB cable between Arduino & computer (PC).
- For interfacing Gas sensor to Arduino, connect A0 pin of Arduino to I/O1 pin of 4 pin connector. Then connect D2 pin of Arduino to Buzzer input.
- Connect the 4 pin connector of Arduino section to 4 pin connector of Gas sensor section.
- If you want to see the sensor data on serial monitor, open the serial monitor of Arduino IDE.
- If you want to see the sensor data on LCD, connect the second 4 pin connector of Arduino section to LCD section. Then connect the A4 pin of Arduino to I/O3 & A5 pin of Arduino to I/O4 pins.
- Observe the output.

Program:**Using Serial Communication -**

```
#include <SoftwareSerial.h>
```

```
int buzzer = 2;
```

```
void setup()
```

```
{
```

```
Serial.begin(9600);
```

```
pinMode(buzzer, OUTPUT);
```

```
delay(1000);
```

```
}
```

```
void loop()
```

```
{
```

```
int mq2_sensor_data = analogRead(A0);
```

```
Serial.println(mq2_sensor_data);
```

```
if (mq2_sensor_data > 300) {
```

```
    Serial.println("Gas leakage");
```

```
    digitalWrite(buzzer, HIGH);
```

```
}
```

```
else {
```

```
    Serial.println("No Gas leakage");
```

```
    digitalWrite(buzzer, LOW);
```

```
}
```

```
delay(500);
```

```
}
```





Result: Study of interfacing Gas sensor to Arduino is successfully studied.

Teacher's Sign.

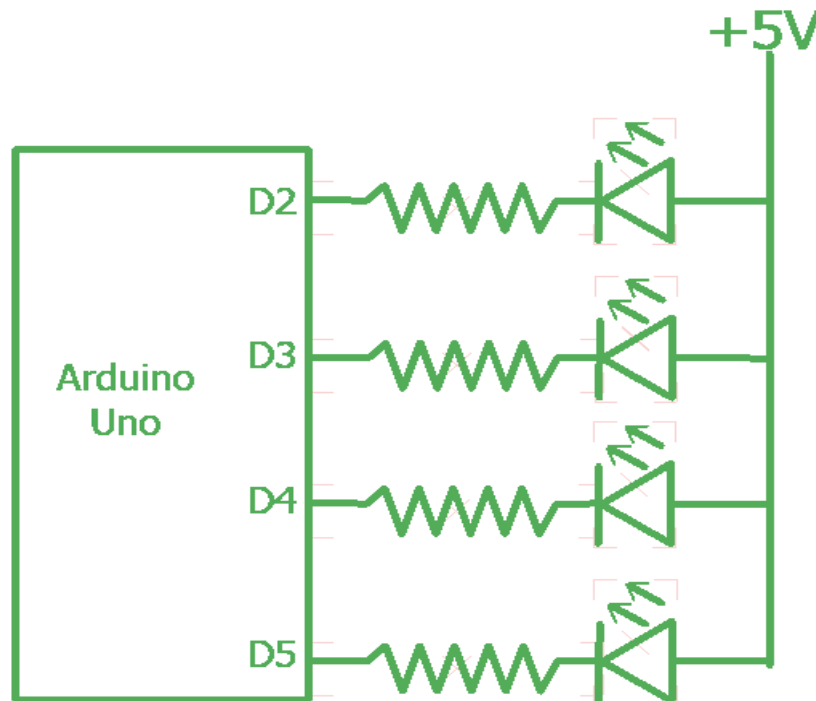
Experiment – 12

Title: To study Arduino based LED pattern generation

Aim: To study interfacing of LED array to Arduino for various pattern generation.

Software Used: Arduino IDE

Circuit Diagram:



Theory:

LED (Light Emitting Diode): LED is a two-lead semiconductor light source. It is a p-n junction diode, which emits light when activated. When a suitable voltage is applied to the leads, electrons are able to recombine with electron holes within the device, releasing energy in the form of photons. This effect is called electroluminescence, and the color of the light (corresponding to the energy of the photon) is determined by the energy band gap of the semiconductor. The charge carriers recombine in a forward-biased P-N junction as the electrons cross from the N-region and recombine with the holes existing in the P-region. Free electrons are in the conduction band of energy levels, while holes are in the valence energy band. Thus the energy level of the holes will be lesser than the energy levels of the electrons. Some portion of the energy must be dissipated in order to recombine the electrons and the holes. This energy is emitted in the form of heat and light. The electrons dissipate energy in the form of heat for silicon and germanium diodes but in gallium arsenide phosphide (GaAsP) and gallium phosphide (GaP) semiconductors, the electrons dissipate energy by emitting photons. If the semiconductor is translucent, the junction becomes the source of light as it is emitted, thus becoming a light-emitting diode, but when

the junction is reverse biased no light will be produced by the LED and, on the contrary, the device may also be damaged.

Program:

[Note – LED's are connected in common anode mode hence 0 = LED ON, 1 = LED OFF]

Pattern 1 (Blinking LED's) -

```
void setup() {  
  pinMode(2, OUTPUT);  
  pinMode(3, OUTPUT);  
  pinMode(4, OUTPUT);  
  pinMode(5, OUTPUT);  
}  
void loop() {  
  digitalWrite(2, LOW);  
  digitalWrite(3, LOW);  
  digitalWrite(4, LOW);  
  digitalWrite(5, LOW);  
  delay(1000);  
  digitalWrite(2, HIGH);  
  digitalWrite(3, HIGH);  
  digitalWrite(4, HIGH);  
  digitalWrite(5, HIGH);  
  delay(1000);  
}
```



Pattern 2 (Alternate LED's) -

```
void setup() {  
  pinMode(2, OUTPUT);  
  pinMode(3, OUTPUT);  
  pinMode(4, OUTPUT);  
  pinMode(5, OUTPUT);  
}  
void loop() {  
  digitalWrite(2, LOW);  
  digitalWrite(3, HIGH);  
  digitalWrite(4, LOW);
```

```
digitalWrite(5, HIGH);  
delay(1000);  
digitalWrite(2, HIGH);  
digitalWrite(3, LOW);  
digitalWrite(4, HIGH);  
digitalWrite(5, LOW);  
delay(1000);  
}
```

Pattern 3 -

```
void setup() {  
  pinMode(2, OUTPUT);  
  pinMode(3, OUTPUT);  
  pinMode(4, OUTPUT);  
  pinMode(5, OUTPUT);  
}  
void loop() {  
  digitalWrite(2, LOW);  
  digitalWrite(3, LOW);  
  digitalWrite(4, HIGH);  
  digitalWrite(5, HIGH);  
  delay(1000);  
  digitalWrite(2, HIGH);  
  digitalWrite(3, HIGH);  
  digitalWrite(4, LOW);  
  digitalWrite(5, LOW);  
  delay(1000);  
}
```

**Pattern 4 (Running LED's) -**

```
void setup() {  
  pinMode(2, OUTPUT);  
  pinMode(3, OUTPUT);  
  pinMode(4, OUTPUT);  
  pinMode(5, OUTPUT);  
}  
void loop() {
```



```
digitalWrite(2, LOW);  
digitalWrite(3, HIGH);  
digitalWrite(4, HIGH);  
digitalWrite(5, HIGH);  
delay(1000);  
digitalWrite(2, HIGH);  
digitalWrite(3, LOW);  
digitalWrite(4, HIGH);  
digitalWrite(5, HIGH);  
delay(1000);  
digitalWrite(2, HIGH);  
digitalWrite(3, HIGH);  
digitalWrite(4, LOW);  
digitalWrite(5, HIGH);  
delay(1000);  
digitalWrite(2, HIGH);  
digitalWrite(3, HIGH);  
digitalWrite(4, HIGH);  
digitalWrite(5, LOW);  
delay(1000);  
}
```



Result: study of interfacing of LED array to Arduino for various pattern generation is successfully studied.

Teacher's Sign.



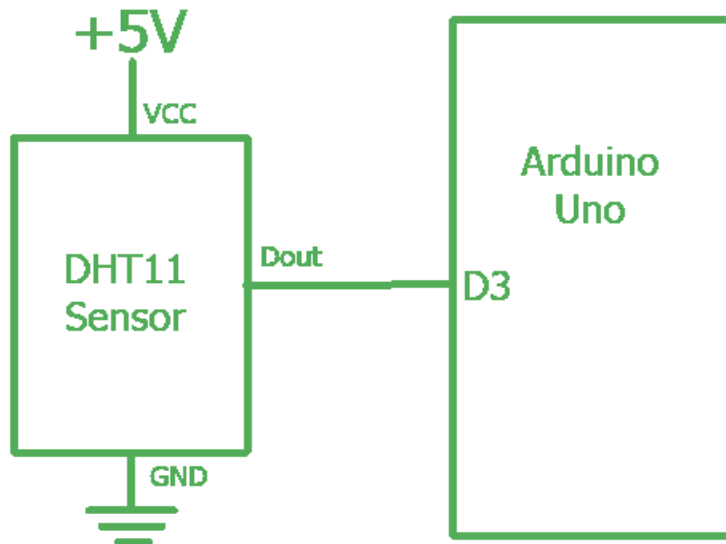
Experiment – 13

Title: Temperature & humidity sensing and monitoring using Arduino

Aim: To study interfacing of Temperature and humidity (DHT11) sensor to Arduino.

Software Used: Arduino IDE

Circuit Diagram:



Theory:

The DHT11 sensor. It is a temperature and humidity sensor having 4 pin single row pin package, single wire serial interface with a calibrated digital signal output. It ensures high reliability, long term stability, excellent quality, fast response, cost effectiveness. The single wire serial interface makes system integration quick and easy. Its small size and low power consumption. It has humidity measurement range of 20% to 90% RH and temperature measurement range from 0 to 50 Celsius.

Procedure:

- Connect the USB cable between Arduino & computer (PC).
- For interfacing DHT11 sensor to Arduino, connect D3 pin of Arduino to I/O1 pin of 4 pin connector.
- Connect the 4 pin connector of Arduino section to 4 pin connector of DHT11 sensor section.
- If you want to see the sensor data on serial monitor, open the serial monitor of Arduino IDE.
- If you want to see the sensor data on LCD, connect the second 4 pin connector of Arduino section to LCD section. Then connect the A4 pin of Arduino to I/O3 & A5 pin of Arduino to I/O4 pins.
- Observe the output.

Program:**Using Serial Communication -**

```
#include<dht.h>

dht DHT;

#define DHT11_PIN 3

void setup() {
  Serial.begin(9600);
}

void loop() {
  int chk = DHT.read11(DHT11_PIN);
  Serial.println(" Humidity ");
  Serial.println(DHT.humidity, 1);
  Serial.println(" Temperature ");
  Serial.println(DHT.temperature, 1);
  delay(1000);
}
```



Result: Study of interfacing Temperature and humidity sensor to Arduino is successfully studied.

Teacher's Sign.

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